nature communications

Article

https://doi.org/10.1038/s41467-026-70660-4

Dual-axis myelination covariance drives the functional connectivity emergence during infancy

Received: 19 August 2025

Accepted: 2 March 2026

Published online: 19 March 2026

Check for updates

Weijin Liu[1,2,3,4], Yuanyuan Chen[1,2,3,4,8] ✉, Xinhao Wang[1,2,3,4], Tianyu Fang[1,2,3,4], Rong Wang[1,2,3,4], Yue Cheng[5], Xin Zhao[1,2,3,4], Qiuyun Fan[1,2,3,4], Wei Gao[6,7] & Dong Ming[1,2,3,4,8] ✉

The mechanisms linking structural maturation to the emergence of functional networks in the perinatal brain remain unresolved. While prevailing models attribute functional connectivity to white matter myelination, neonates paradoxically exhibit adult-like resting-state networks despite profoundly immature white matter tracts. Here, we proposed gray matter myelination covariance as a critical basis of early functional connectivity emergence. We introduced a dual-axis myelination covariance framework and derived a myelination-function coupling (MFC) index specific to the newborn brain. Results revealed that the MFC exhibited distinct spatial patterns dominated by primary sensory and motor cortices, increased with age, and showed a distance-dependent strength. Crucially, neonatal MFC patterns showed a strong spatial correlation with gene expression profiles implicated in neurovascular coupling and specifically predicted later behaviors. These findings suggest that during infancy, the integration of brain function is not initially dominated by only the white matter connections but is also shaped by the synchrony of intracortical microstructure that reflects shared developmental trajectories, which offers a framework for understanding the formation of the developmental connectome.

The perinatal period and early postnatal weeks constitute a critical epoch for cerebral development, characterized by orchestrated multidimensional reorganization of neural architecture and emergent functional networks[1–4]. During this phase, rapid cortical expansion is accompanied by explosive synaptogenesis, dendritic arborization, and myelination processes that collectively scaffold the nascent functional networks observed in neonates by 37-44 weeks of gestational age[5–7]. Since the first adult-like default-mode functional network was identified in neonates' brain[4], a hierarchical process of early brain functional network development has been delineated over the past two decades[3,8]. Functional networks develop through hierarchical circuit assembly: starting before birth, primary sensory/motor integration stabilizes by 3 months of age, limbic-cortical regulation emerges for initial emotional control by 24 months, and advanced associative networks like the default-mode network begin crystallizing after 2 years of age[3,7]. However, the underlying mechanisms linking discrete functional capacities with structural elements, in particular myelination, which establishes the biophysical substrate for signal propagation, remain elusive.

[1]Medical School, Faculty of Medicine, Tianjin University, Tianjin, China. [2]State Key Laboratory of Advanced Medical Materials and Devices, Tianjin University, Tianjin, China. [3]Haihe Laboratory of Brain-Computer Interaction and Human-Machine Integration, Tianjin, China. [4]Tianjin Key Laboratory of Brain Science and Neuroengineering, Tianjin, China. [5]Department of Radiology, Tianjin First Center Hospital, Tianjin, China. [6]Cedars–Sinai Biomedical Imaging Research Institute, Los Angeles, CA, USA. [7]Departments of Biomedical Sciences and Imaging, Cedars-Sinai Medical Center, Los Angeles, CA, USA. [8]These authors jointly supervised this work: Yuanyuan Chen, Dong Ming ✉e-mail: bain@tju.edu.cn; richardming@tju.edu.cn

The coupling mechanisms[9,10] of brain structure and function are key to understanding early brain functional integration. Current efforts to map structure-function coupling in the neonatal brain reveal fundamental paradoxes that challenge conventional neurodevelopmental frameworks. While adult studies demonstrate moderate spatial correspondence between structural connectivity and functional connectivity, neonatal data display that long-range connections often exhibit weaker structure-function alignment (lower coupling)[10–13]. This characteristic is interpreted as a functional specification of the cerebral cortex during the early development[12,14–16], allowing these regions to display more flexible functional connectivity that is not strictly constrained by physical axonal wiring. The distance-dependent decoupling mechanism contradicts the hypothesis that functional networks emerge from progressive myelination of long-distance white matter tracts[8]. More notably, the age-dependent decoupling phenomenon of brain structure and function during early development[10] also contradicts other studies reporting that the structural-functional coupling undergoes sustained enhancement from 30 weeks of gestational age through approximately 20 years of age[9,14,17]. This increased coupling is driven by the parallel growth of both integration and segregation between the two connectomes. However, an important neural structural basis, myelination, in the infant brain is often overlooked. Myelination is known to be activity-dependent, thus coordinated developments with functional connectivity are expected[18–22]. However, there is currently no previous study examining the direct, quantitative relationships between coordinated myelination and functional synchrony in the developing brain. Moreover, although white matter myelination has been more extensively studied, it remains unresolved whether it is the myelination of long-range tracts or the maturation of local gray matter that correlates more closely with the emergent functional connectivity. These represent critical knowledge gaps in better understanding infant structural-functional development.

As extensively reported, myelination is immature in the newborn stage. However, despite far-from-mature myelination levels in white matter tracts[23,24] (30% of the adult level), neonates exhibit resting-state functional networks sharing 80% spatial similarity with adult counterparts[25,26]. This suggests that white matter-based structure-function coupling is likely insufficient to fully explain the underlying dynamics of the developing infant brain. Therefore, we hypothesize that the formation of functional connectivity does not solely rely on white matter fiber connections. Instead, we propose that coordinated growth of local gray matter myelination is more closely related to the emergence of initial functional connectivity. The brain wiring, including synaptic formation and pruning within the signaling-integrated white matter, is more potently modulated by the distributed gray matter systems[27–31]. Notably, gray matter, which primarily consists of neuronal cell bodies, develops earlier in the fetus than white matter, and undergoes rapid growth in late gestation and continues to develop after birth[5–7]. Unlike white matter myelination that peaks postnatally, gray matter oligodendrogenesis commences as early as 28 weeks of gestational age, synchronizing with the emergence of local field potential oscillations[1,27,28,32,33]. Such findings imply that early functional architectures may arise from the cumulative cortical growth.

Homophilic principles of nervous system organization state that brain regions with similar developmental trajectories will be similar also in terms of other properties[33–35], hypothetically owing to shared genetic and/or convergent environmental effects. The cerebral cortex undergoes a process of spatially organized and heterochronic maturation[34,36]. Structural covariant connectivity characterizes the coordination of neural development measured by statistical similarity in developmental trajectories or distributions of structural indicators between paired brain regions. Over the past two decades, various methods of structural covariance analysis have been developed[37–40], primarily encompassing both population-level and individual-level dimensions, and were used to study heterochronic network development[41,42]. Emerging evidence implicates gray matter myelination as a pivotal yet understudied mediator[18,27,43,44]. Cortical myelination is correlated with cortical development and intracortical circuit complexity[45,46], and gray and white matter myelination are significantly correlated[47]. The first wave of myelination initiates prenatally, during which the primary sensory and motor cortices undergo myelination, and heteromodal association cortices do not become fully myelinated until the second decade, which is similar to the hierarchical development of cortical function[5,18,48]. Additionally, the structural covariance connectome also has similar dense characters[49] with the functional connectome that even strong connectivity could be performed without direct fiber connectivity. Therefore, we held the hypothesis that the homophilic cortical growth measured by the myelination covariance could support the emergence of functional synchronization during infancy.

This study proposed a dual-axis myelination covariance framework to quantify the homophilic development, which encodes how regionally heterogeneous myelination schedules co-vary across both the cohort and neuroanatomical space, captured by two dimensions of covariance: (1) inter-subject covariance calculated as the group-level myelination covariance (gMC), representing synchronized myelination trajectories across subjects, and (2) intra-subject covariance calculated as subject-level myelination covariance (sMC), mapping coordinated myelination stages between distinct brain areas within individuals. This dual-axis covariance framework provides the statistical foundation for investigating spatially coordinated myelin growth that may show activity-dependent growth with emergent functional connectivity. Based on the dual-axis myelination covariance computed from myelin-sensitive MRI metric (T1w/T2w ratio), we further proposed a structure-function coupling approach called myelination-function coupling (MFC), which is specific for newborn brain development (Fig. 1) and is superior to traditional white matter-based structure-function analyses. A distinct MFC pattern was identified in neonates with multimodal brain imaging data from the Developing Human Connectome Project (dHCP)[50]. The single axis myelination covariance, either sMC or gMC, was also derived sMFC or gMFC to compare. Then, this study explored several factors associated with the MFC, including the age effects, the distance dependence, the birth effects, the transcriptomics bases, and the behavioral association. Results indicated that this coupling index can unveil the distinct features of how the brain's functional organization emerges from its parallel structural basis during the newborn stage.

Results

The dominant myelination-function coupling in the sensory cortex

We first examined the presence of association between functional connectivity (FC) and myelination covariance connectivity (MC) at both group and individual levels (Methods: Calculation of myelination-function coupling) using the public multi-modal imaging data for full-term neonates from the dHCP (Methods: Participants and data acquisition). The group level whole-brain connections of the two single-axis myelination covariance matrices were both significantly aligned with functional connectivity (gMC for the intra-hemisphere/whole connectome: $R^2 = 0.30/0.23$, $\beta = 0.45/0.38$, $p < 0.0001/p < 0.0001$, sMC for the intra-hemisphere/whole connectome: $R^2 = 0.22/0.23$, $\beta = 0.097/0.087$, $p < 0.0001/p < 0.0001$; Fig. 2a), which were consistent to the inter-hemisphere connections (see Supplementary Fig. 1a). The averaged maps of individual MFC, as well as those of gMFC and sMFC, were illustrated (Fig. 2b, c), showing an expanded higher MFC level than either gMFC or sMFC. Dominant regions in the MFC map included the primary motor cortex, posterior insular cortex, lateral temporal cortex, and primary visual cortex. Distinct regions—such as the primary auditory cortex and primary visual cortex—were dominant for gMFC, whereas regions including the primary motor cortex, insula cortex and

https://doi.org/10.1038/s41467-026-70660-4

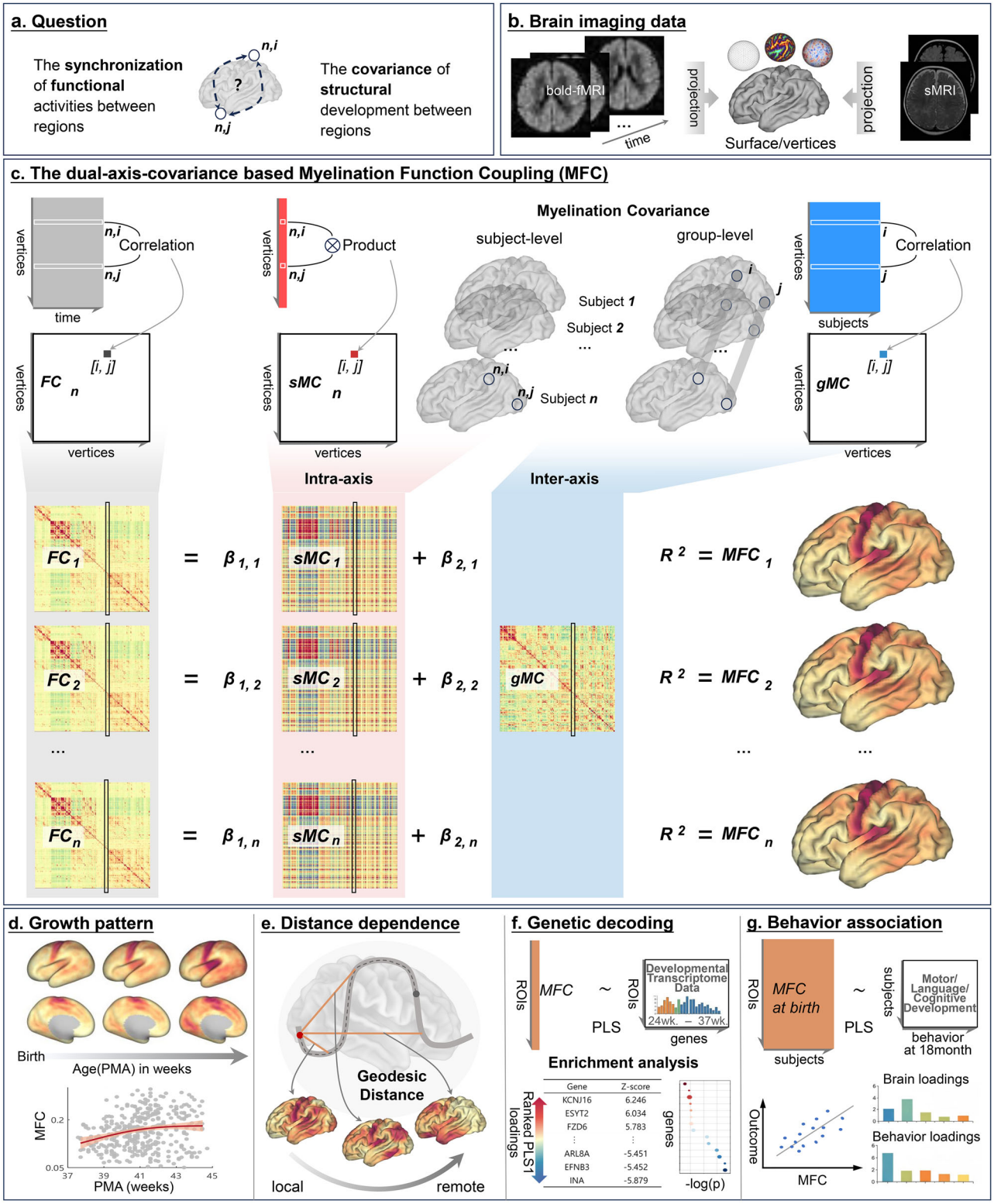

Fig. 1 | The research pipeline of this study. a Basic question of this study. **b** The brain imaging data and projection to the cortical surface. **c** The proposed workflow of myelination-function coupling (MFC) based on dual-axis covariance. **d** Developmental patterns of the MFC index. **e** Spatial distance dependence analysis of the MFC map. **f** Genetic decoding using transcriptional enrichment analysis. **g** The association analysis between behavior outcomes at 18-month-age and brain MFC at birth. MATLAB R2022b (MathWorks, Natick, MA, USA) and Connectome Workbench were used for visualizations.

https://doi.org/10.1038/s41467-026-70660-4

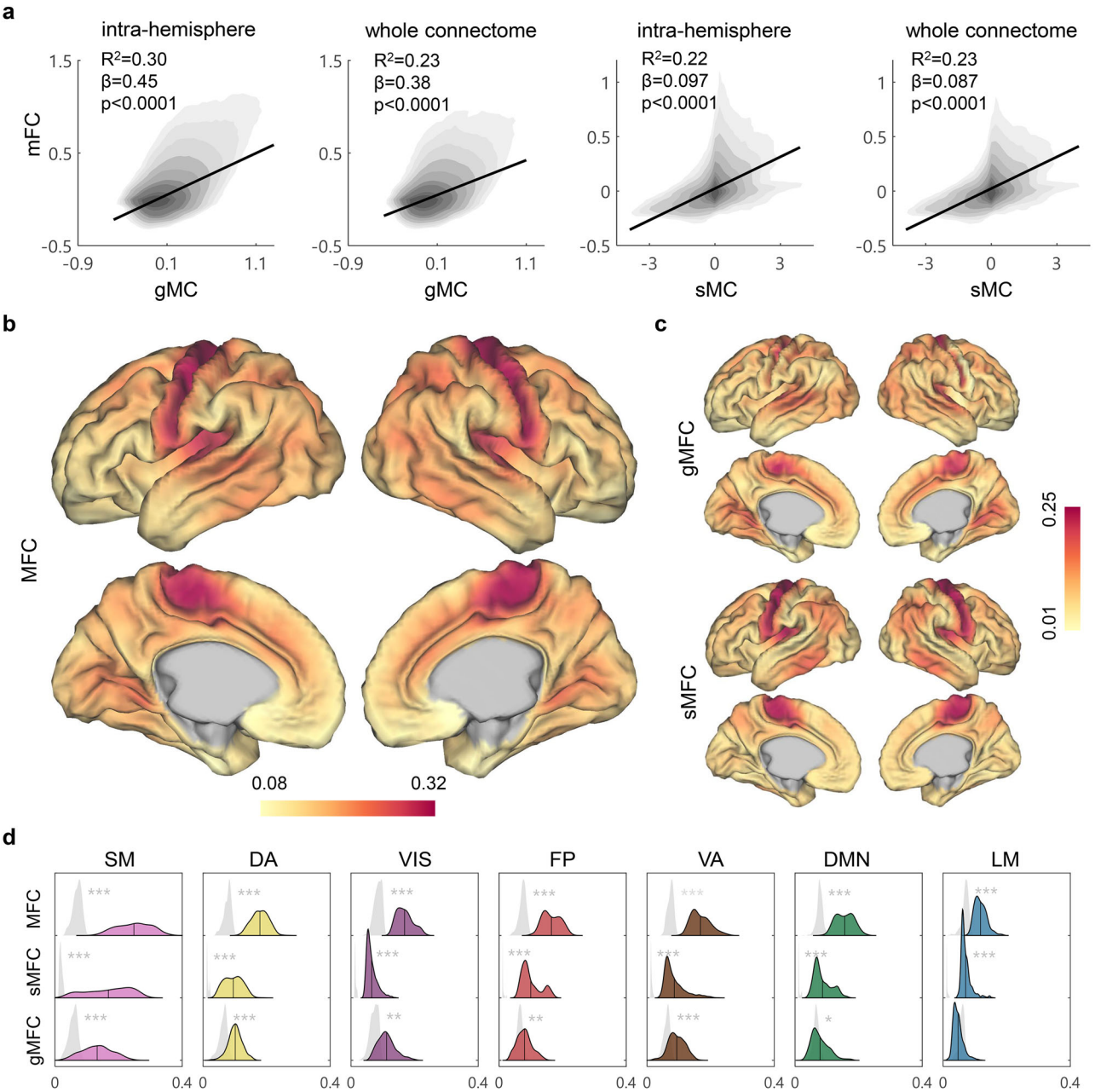

Fig. 2 | Spatial pattern of myelination-function coupling index. a Linear regression between mean functional connectivity (mFC) and group-level myelination covariance (gMC) or subject-level myelination covariance (sMC) for the connections of intra-hemisphere or whole connectome (n = 18,438,448 for hemisphere; n = 36,881,166 for whole brain), with the background displaying a density distribution map. The exact p equals zero for each of the four associations. **b** The spatial pattern of the group-averaged MFC with dual-axis myelination covariance. **c** The spatial patterns of gMFC and sMFC. gMFC: the group-level MFC derived from dual-axis framework only using gMC; sMFC: the subject-level MFC derived from dual-axis framework only using sMC. **d** System-level distributions of the three coupling indices (displayed in colored histograms for each system), most of which are significantly different from the null distribution (displayed as grayscale histograms), and vertical lines in the graph represent the mean values. * $p_{spin} < 0.05$; ** $p_{spin} < 0.01$; *** $p_{spin} < 0.001$. The investigated systems are visual (VIS), sensorimotor (SM), frontoparietal (FP), default mode (DMN), limbic (LIM), dorsal attention (DA), and ventral attention (VA) systems. Source data are provided as a Source Data file. Analyses and visualization were performed using custom scripts in MATLAB R2022b (MathWorks, Natick, MA, USA) together with Connectome Workbench.

inferior temporal cortex were mainly dominant for sMFC. Among the three coupling indices, gMFC and sMFC exhibited lower similarity (corr = 0.38, $p < 0.0001$), which was substantially lower than their similarities with MFC (corr = 0.78 for gMFC and 0.83 for sMFC, both $p < 0.0001$). Spin permutation tests on the three coupling indices revealed that the majority of vertices had significantly higher MFC than the null coupling level following 5000 randomizations (p_{spin} <0.05; Supplementary Fig. 1b). Moreover, the coupling indices at the system level (Yeo's 7-network parcellation[51]) also showed significance (Fig. 2d and Supplementary Table 1). The differences observed in mean values among the three coupling indices were further confirmed by independent two-sample t-tests at the functional system level (Supplementary Table 2).

To further investigate the spatial hierarchy of MFC, we computed the principal gradient of the FC connectome and correlated it with the MFC pattern. High consistency was observed (corr = 0.69 at Icosahedron-1002 atlas level, $p < 0.0001$; Supplementary Fig. 1c; corr = 0.67 at vertex level, $p < 0.0001$), suggesting that the MFC hierarchical

pattern also captured the regions' functional gradient variance. In addition, the proposed dual-axis MFC framework was repeated at the atlas level, yielding atlas-level MFC values using two alternative approaches for individual structural covariant connectome: KL divergence-based sMC (Kullback-Leibler divergence[52,53] of probability distribution of myelin between paired brain regions) and the atlas-level sMC based on dot product proposed in this study. Using neonatal AAL atlas, comparisons among the vertex-level MFC map (after averaging to the atlas level) and the two atlas-level MFC maps revealed highly similar spatial patterns overall (corr = 0.52, $p < 0.0001$ for KL divergence-based MFC, and corr = 0.61, $p < 0.0001$ for dot-product-based atlas-level MFC, compared with the atlas-mean MFC; Supplementary Fig. 2a), with higher similarity observed between the two atlas-level MFCs (corr = 0.81, $p < 0.0001$). Using Icosahedron-1002 atlas with fine-granularity, a more consistent MFC pattern with the vertex-level was further observed (corr = 0.98, $p < 0.0001$ for vertex-level MFC and atlas-level MFC; Supplementary Fig. 2b). Considering the subcortical regions, the MFCs were highly consistent (corr = 0.98, $p < 0.001$) between that eighter with or without the subcortical regions in the calculating framework (Supplementary Fig. 3 and Supplementary Result 1). Furthermore, a direct coupling between the functional connectivity profile and the myelin map showed significantly lower coupling than our proposed MFC (see Supplementary Fig. 4 and Supplementary Result 2).

Limited contribution from the white matter connectivity for the coupling

Having established that structural covariance of gray matter myelination is associated with the emergence of functional connectivity, it remained possible that white matter fiber connections also play a role therein. Therefore, we incorporated diffusion tensor imaging (DTI) data for fiber tract reconstruction, calculated the fiber tract connectivity density between brain regions to construct the structural connectivity network (SCN) based on the UNC infant AAL atlas. Subsequently, the fully weighted SCN was used to model the coupling relationship between white matter tracts and functional connectivity. The coupling was calculated using the same method as MFC, adopting a linear regression model. Averaged map of atlas-level individual MFC and SFC were illustrated (Supplementary Fig. 5a, b), showing that a moderate level of correlation between them (corr = 0.48, $p < 0.0001$; Supplementary Fig.5c). MFC map showed much higher level than SFC across the whole brain (the mean MFC is 0.22, and the mean SFC is 0.09), except for the cuneate cortex, with the peak increment up to 9 times (Supplementary Fig. 5d, the coupling increment map of MFC over SFC). The dominant regions of SFC were only distributed in the medial cortex, including the primary visual cortex and primary motor cortex. Furthermore, we introduced structural connectivity (SC) as an additional predictive variable into the MFC dual-axis framework. For the majority of brain regions, the introduction of SC profile barely improved the coupling strength, particularly in the dorsolateral prefrontal cortex (Supplementary Fig. 5e, MFC map with SC included). Regions with an increment exceeding 20% were concentrated in the posterior cingulate cortex, calcarine cortex, and cuneus (Supplementary Fig. 5f, Coupling increment map of SC-included MFC over MFC without SC variable).

The spatial growth patterns of myelination-function coupling

To identify the typical development trajectories of MFC during the neonatal period (Fig. 3a), we applied a generalized additive model (GAM) to capture both linear and non-linear relationships between MFC and postmenstrual age at scan (in weeks). Vertex-wise GAM for MFC demonstrated broadly distributed significant growth, including the whole visual cortex, inferior temporal cortex, parieto-occipital junction, sensorimotor cortex, insula, and prefrontal cortex ($p_{FDR} < 0.05$; Fig. 3b). Partially, the sensorimotor cortex, insula cortex, temporal cortex, and parieto-occipital junction exhibited significance for sMFC, whereas only the visual cortex was significant for gMFC (Fig. 3c). Notably, the primary motor cortex with dominant MFC at the vertex-level was not dominant in this growth pattern. Significant growth of whole-brain averaged MFC was also identified using GAM, with similar coefficients for sMFC and gMFC ($\beta_{growth\ rate} = 0.0049$, $p < 0.0001$ for both; Supplementary Fig. 6a), and a higher coefficient for MFC ($\beta_{growth\ rate} = 0.0083$, $p < 0.0001$). When parcellating the whole brain into seven systems and applying locally weighted scatterplot smoothing (LOWESS), the three coupling indices showed parallel increasing across all seven systems (Supplementary Fig. 6b), with the highest values in the sensorimotor system. Notably, sMFC and MFC showed a higher growth slope in the visual cortex than gMFC. The linear model was repeated for the global-level and vertex-level coupling indices to assess stable linear effects (Supplementary Fig. 6c, d).

The growth-rate coefficients were delineated for each system, with the highest values in the visual and motor systems, and MFC exhibited the highest coefficients across all systems except the sensorimotor system, relative to sMFC and gMFC (Fig. 3d). Coefficient comparisons showed that sMFC was significantly higher than gMFC in the sensorimotor (SM), frontoparietal (FP), default mode (DMN), limbic (LIM) and ventral attention (VA) systems, while gMFC was higher in the visual (VIS) and dorsal attention (DA) systems (Fig. 3d and Supplementary Table 3). The spider plot (Fig. 3e) illustrated the percentage of significantly growing regions per system, indicating complementary distributions of sMFC and gMFC growth across the seven systems, and that MFC could capture all such growth effects.

The distance dependence of myelination-function coupling

Considering distance-related wiring costs during functional network emergence and potential overestimates of the coupling between myelination and functional profile caused by the spatially adjacent vertices, the cortical distance dependence of myelination-function coupling was further explored. The vertex-to-vertex connection profiles were ranked according to the geodesic distance between vertices and equally divided into multiple subsets from local to remote connections. Then, a series of MFC maps was calculated. The MFC spatial patterns exhibited high dynamics from the local to the remote connections, and each subset of connections showed distinct spatial MFC pattern (nine subsets, about 11% per range for each; Fig. 4a). Regions including the primary motor, primary visual and auditory cortex displayed dominant MFC for short-range connections, whereas the inferior temporal lobe and medial prefrontal cortex showed high MFC for long-range connections. Using finer range divisions (fifty subsets, 2% per range; the similarities between MFC maps of paired subsets in Supplementary Fig. 7a), the distance-dependent MFC curves within each system were further illustrated (solid curves in Fig. 4b), showing significantly higher MFC (around 0.4) for the most local connections and relatively lower values (around 0.15) for other distance ranges. Notably, the distance-dependent decrease in MFC exhibited an inflection point at around 12% percentile across all systems. Prior to this point, MFC decreased rapidly and linearly, whereas after this point, MFC varied smoothly and nonlinearly from local to remote connections.

Leveraging these distance-dependent subsets, we further generated distance-dependent linear growth rate curves for each system (see dashed curves in Fig. 4b), showing a gradual increase from local connections (growth rate consistently around zero) to remote connections, with distinct patterns for each system. Linear relationships between the growth rate and geodesic distance were mapped as coefficients on the cortical surface (Fig. 4c). Positive regions indicated higher growth rate for long-range connections, as exemplified by distinct regions including the middle temporal gyrus, the premotor cortex and supplementary motor area (Fig. 4c). Negative regions indicated higher growth rate for short-range connections, as exemplified by the distinct regions including the primary motor, the

https://doi.org/10.1038/s41467-026-70660-4

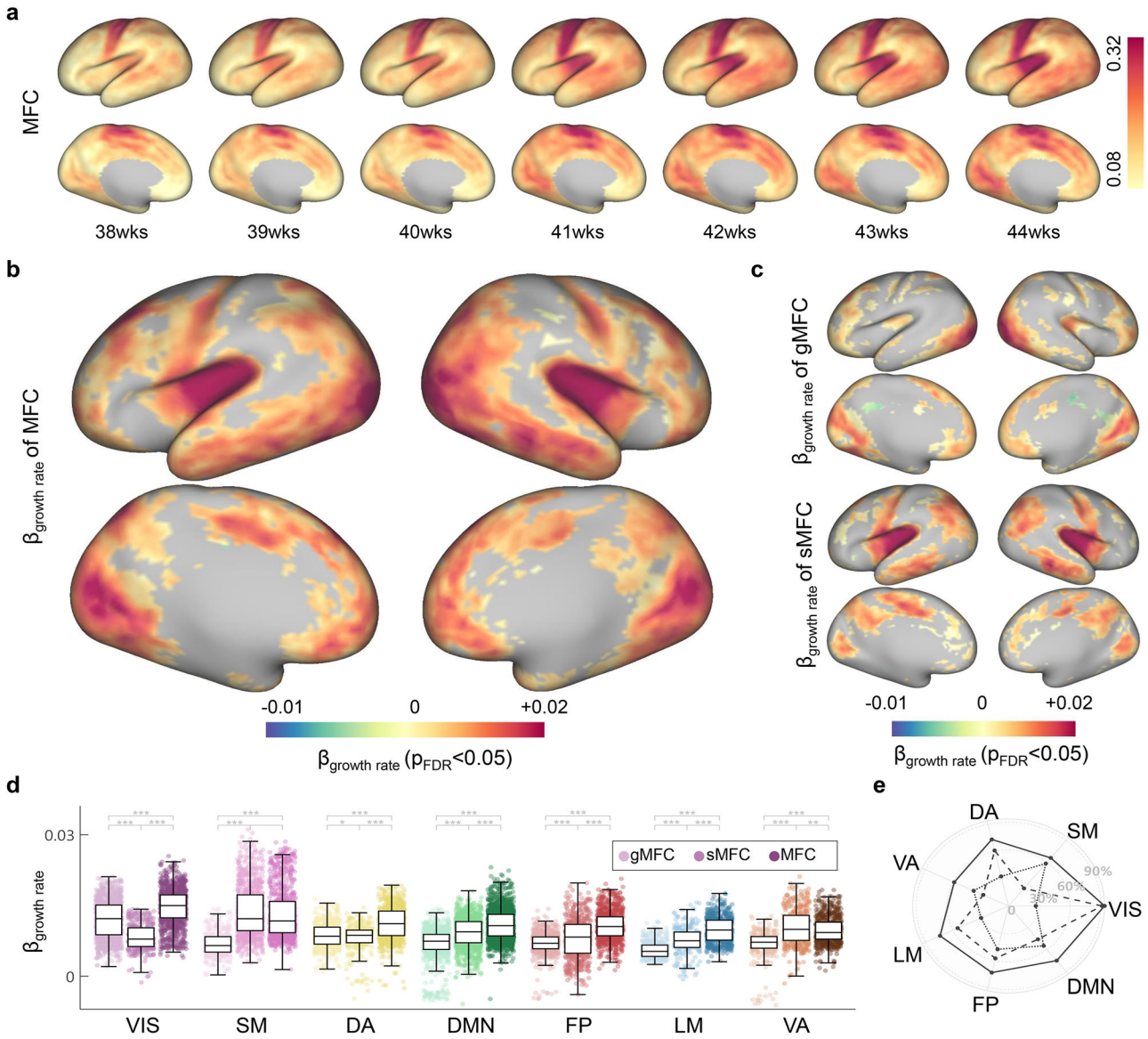

Fig. 3 | Typical development of the myelination-function coupling. a The spatial-temporal maps of MFC for a series of PMA windows. **b** Growth-rate coefficient of MFC during infancy derived from GAM analysis with $p_{FDR} < 0.05$. **c** Growth-rate coefficient of gMFC and sMFC with $p_{FDR} < 0.05$. **d** The distributions and comparisons of the growth-rate coefficient at significant vertices in GAM of the three coupling indices, and the statistical differences between them are all significant (* $p_{FDR} < 0.05$; ** $p_{FDR} < 0.01$; *** $p_{FDR} < 0.001$, the number of significant vertices for each system and each index is listed in Supplementary Table 4). Boxes display the interquartile range (IQR: lower hinge: 25th percentile; upper hinge: 75th percentile; center line: median). **e** Radar chart demonstrates the proportion of significantly developed areas for each system, with solid line for MFC, large dashed lines for gMFC and small dashed lines for sMFC. The investigated systems are visual (VIS), sensorimotor (SM), frontoparietal (FP), default mode (DMN), limbic (LIM), dorsal attention (DA), ventral attention (VA) systems. Source data are provided as a Source Data file. Analyses and visualization were performed using custom scripts in MATLAB R2022b (MathWorks, Natick, MA, USA) together with Connectome Workbench.

posterior insula and the inferior parietal lobule (Fig. 4c). To evaluate the potential distance effect on the MFC patterns, we updated the MFC maps using two approaches: distance profile regressing (Fig.4d) and local connections removing (Fig.4e). Both maps showed high spatial similarity to the original MFC map, with significant correlations (corr = 0.86 and 0.88, $p < 0.0001$; Supplementary Fig. 7b).

Similar distance-dependent MFC patterns were also reproduced across the five cyto-architectonic classes (Supplementary Fig. 7c) and for several critical regions (Supplementary Fig. 8). Convergently in details, the MFC of VIS and DA exhibited higher growth sensitivity (highest growth rate) to the medium distance connections (Fig. 4b), which was similar to that of the polar (POL) and parietal (PAR) cyto-architectonic classes; the MFC growth rate of DMN was lower for medium distance connections but higher for both local and remote connections; the MFC growth rate of VA showed higher for remote connections, which was consistent with the agranular (AGR) and granular (GRA) cyto-architectonic classes; the MFC growth rate of FP was higher for local connections, a pattern shared with the PAR cyto-architectonic class. To further validate our results, we replicated key findings using the updated MFC with distance regressed (see Supplementary Fig. 9). These replicated analyses included: (1) the global association between gMC/sMC and mFC; (2) the similarity between the updated MFCs and the original MFCs for 50 subsets; (3) the distance-dependent MFC curves and growth rate curves; (4) the global MFC growth rate and vertex-wise growth-rate coefficient of MFC with $p_{FDR} < 0.05$.

https://doi.org/10.1038/s41467-026-70660-4

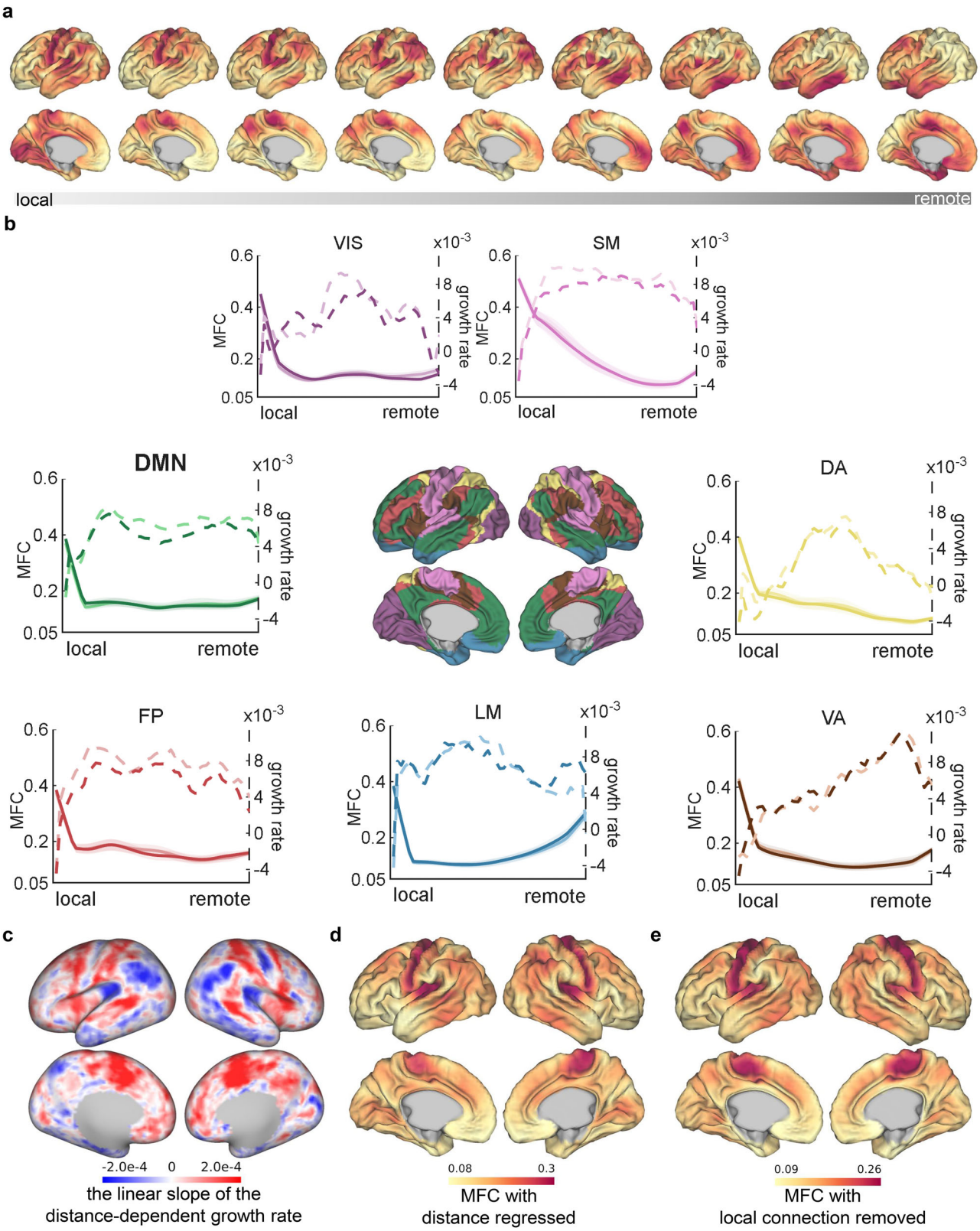

Fig. 4 | Distance dependence of the MFC patterns and the growth rate. a Spatial patterns of MFC from local connections to remote connections. **b** MFC curves (solid lines, with locally weighted scatterplot smoothing) and growth rate curves (dashed lines) against the distance for each system, with the dark colored curves and light colored curves respectively representing the right and left hemispheres respectively. The shadows for solid lines show the upper 60th percentile and the lower 40th percentile from $n = 364$. **c** Linear fitting coefficients at each vertex between the growth rate of MFC and cortical distance, positive values represent a higher growth rate for long-distance coupling, negative values represent a lower growth rate for short-distance coupling. **d** Averaged map of individual MFC with distance regressed. **e** Averaged map of individual MFC with local connections removed. Source data are provided as a Source Data file. Analyses and visualization were performed using custom scripts in MATLAB R2022b (MathWorks, Natick, MA, USA) together with Connectome Workbench.

The specific effect of birth on myelination-function coupling

Next, we sought to explore the differential influences of gestational age (GA) at birth (defined as the weeks from fertilization to birth) and postnatal age (PNA) at scan (defined as the time from birth to MRI scan) on the development of cortical MFC. These two age metrics reflect the distinct effects of intrauterine and extrauterine environments, considering birth as a critical event involving environmental changes. We compared two prediction models for the vertex-wise MFC using different explanatory variables: (I) only postmenstrual age (PMA) at scan; (II) GA and PNA combined. The adjusted coefficient of determination R^2_{adj} of model II (based on GA and PNA combined) outperformed that of model I (based on PMA alone) (Fig. 5a), with vertices more widely distributed in the high R^2_{adj} value range. Specifically, the mean R^2_{adj} of model II was higher than that of model I: for model II, vertices with $p_{FWE} < 0.05$ account for 95.2%, with a mean R^2_{adj} of 0.0368 (range: 0.002– 0.146) at these significant vertices; for model I, vertices with $p_{FWE} < 0.05$ account for 92.4%, with a mean R^2_{adj} of 0.0306 (range: 0.003–0.140) at these vertices. The model fitness of the two models across different systems was also compared (Fig. 5c and Supplementary Table 7; only vertices with fitting effect meeting $p_{FWE} < 0.05$ were included). Higher R^2_{adj} were observed for Model II (all $p < 10^{-5}$), especially in the SM, DMN, and FP systems.

To exemplify the effect of birth for MFC, we attempted to measure the different developmental patterns of MFC before and after birth, using the linear estimated coefficients of MFC for intrauterine and extrauterine development separately. A positive MFC growth associated with GA was observed across the whole cortex (Fig. 5b), particularly in the motor and visual cortices. For extrauterine development, all regions exhibited positive MFC growth except for the primary motor cortex and the posterior cingulate cortex (Fig. 5b). The coefficients comparison between GA and PNA suggested that the GA effect was significantly greater than PNA effect (all $p < 10^{-5}$; Fig. 5c and Supplementary Table 6), indicating a more profound influence from the intrauterine development. To further verify the birth effect on MFC development, we extracted two participant subsets: (1) full-term neonates with small PNA, representing intrauterine development, (2) preterm neonates with term-equivalent PMA and long PNA, representing extrauterine development (Fig. 5d). Both MFCs showed similar cortical patterns with higher values in the motor cortex. In comparison, full-term neonates exhibited significantly higher MFC levels ($p < 10^{-5}$; Fig. 5e, f), especially in the inferior parietal lobule (IPL), precuneus (PCUN), and superior frontal gyrus (SFG).

The transcriptomic association of myelination-function coupling

To explore the transcriptomic basis of the myelination-function coupling during neonates, this study utilized the BrainSpan Atlas of the Developing Human Brain, aged from 24 post-conceptual weeks (pcw) to 37 pcw with 11 regions. The first component of partial least squares (PLS) depicted gene expression pattern that was mostly associated with MFC distributions, which explained 20.38% of the variance in gene expression data ($p_{perm} = 0$ with 5,000 times permutation tests; Fig. 6a). Meanwhile, under the PLS1 (first component of PLS), significant spatial association between the MFC score and the gene expression score was observed (corr = 0.91, $p = 0.0001$; Fig. 6b). Furthermore, by utilizing a univariate one-sample Z test, we ranked these candidate genes based on normalized PLS1 weights, identifying 352 positively weighted genes (PLS1 + : $Z > 3$, $p_{FDR} < 0.001$; Fig. 6c) and 112 negatively (PLS1-: $Z < -3$, $p_{FDR} < 0.001$; Fig. 6c). We utilized the Metascape[54] to align PLS1+ and PLS1- genes with the gene ontology (GO) terms, including biological process (BP), molecular function (MF) and cellular component (CC). The PLS1+ gene list exhibited significant enrichment in terms related to angiogenesis, transmembrane transport, neural regulation and immune response (the top 20 GO enriched clusters shown in Fig. 6d, orange bars, and Fig. 6e), such as “side of membrane”, “tube morphogenesis”, “transepithelial transport” and “gliogenesis”. The PLS1- genes were associated with neuronal morphogenesis and neural regulation (see the Fig. 6d, blue bars), such as “synaptic membrane”, “neuron projection cytoplasm” and “axon”.

The behavioral association of myelination-function coupling

Behavioral assessments from the Bayley III scale for dHCP subjects at approximately 18 months of age were associated with the individual MFC values in this study. Non-parametric permutation analysis (5000 times permutation) for null-hypothesis testing revealed a statistically significant population correspondence between the regional MFC and behavioral profiles. Based on PLS, whole-brain MFC and behavioral scores for PLS1 showed a significant correlation (adjusted $R^2 = 0.026$, $p_{perm} = 0.006$; Fig. 7a), with the negative behavior loadings and positive image loadings. Similarly, under the PLS1, whole-brain functional connectivity strength (FCS) and behavioral scores also showed significance (adjusted $R^2 = 0.047$, $p_{perm} = 0.001$; Fig. 7b), with corresponding behavior and imaging loadings. For functional system-level PLS analysis, remote-distance sensorimotor MFC was significantly associated with the motor behavioral domain (adjusted $R^2 = 0.045$, $p_{perm} = 0.041$; Fig. 7c). Notably, for the 9 distance subsets (see Fig. 4a), the distance-specific MFC-behavior relationships (adjusted R^2 from scores for PLS1) exhibited increasing association with the connection distance, with adjusted R^2 values increasing from 0.020 for the local connections to 0.092 for the remote connections (Fig. 7d and Supplementary Fig. 10).

Discussion

This study sheds light on a fundamental problem in developmental neuroscience concerning the neonatal brain: how adult-like functional networks emerge despite the profound immaturity of white matter in the neonatal brain. Traditional models of brain development have predominantly emphasized the role of white matter myelination in the maturation of functional connectivity[9,12,14,55]. These models suggest that functional networks emerge during infancy primarily from the myelination of long-distance axonal pathways, with the speed of signal transmission facilitated by white matter serving as a key structural basis for the synchronization of neural activity. However, our findings challenge this hypothesis by showing that, during the newborn stages of brain development, functional connectivity is not solely dependent on the maturation of white matter tracts. Instead, the structural developmental homogeneity of gray matter, particularly the synchronous myelination within cortical regions, plays a pivotal role in shaping early functional network organization. Our proposed approach integrates cortical myelination covariance along two distinct dimensions (inter-subject and intra-subject) with the myelination-function coupling (MFC) calculation, offering a more nuanced insight into the relationship between structure and function in the neonatal brain. Our study presents imaging-based evidence for the coordinated development between regional functional connectivity and myelination covariance. This coordination is predominantly anchored in gray matter with shared neurodevelopmental mechanism[18,24,44], rather than being solely confined to white matter[24,56].

Synchronized myelination stages across cortical regions reflect common developmental trajectories influenced by genetic programming and environmental stimuli. As regions mature in tandem, they develop functional coherence, rendering them more prone to synchronizing their neural activity, even before fully mature axonal pathways establish connections between them[43,57,58]. This process is supported by the role of myelination in neuronal electrophysiological maturation and local circuit refinement. A recent MRI study indicated that the fidelity and power ratio of functional signal transmission across the gray-white matter boundary have a strong positive association with the local myelination[59]. Essentially, myelination covariance may reflect the coordinated maturation of functionally linked neural

https://doi.org/10.1038/s41467-026-70660-4

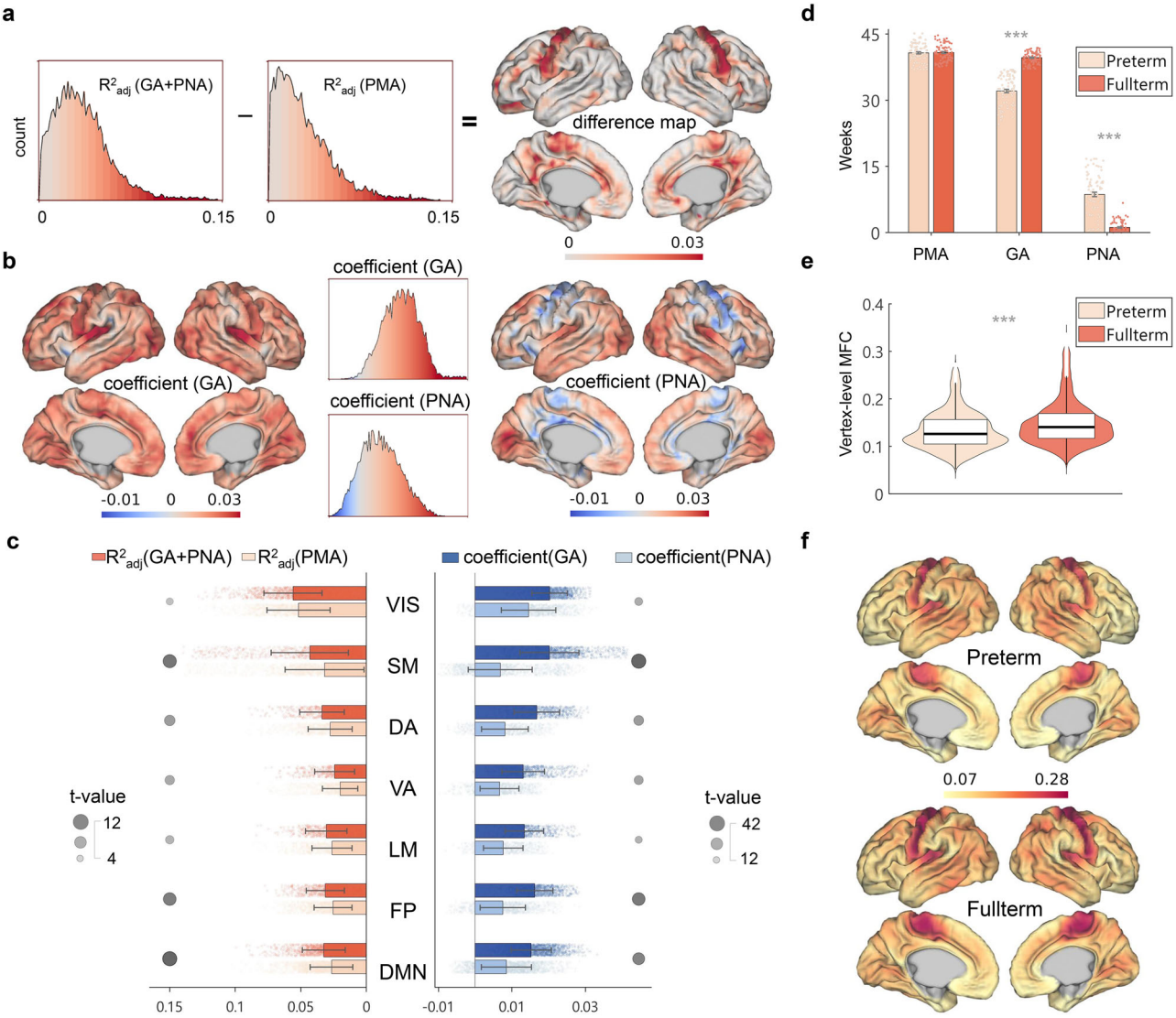

Fig. 5 | Birth effect on the MFC patterns. a Regression performance of model II (GA and PNA combined) is greater than that of model I (only PMA) at the vertex level. **b** Distribution of GA coefficient and PNA coefficient at vertex level. **c** The regression performances (R^2_{adj} of model I and model II in the left panels) or the coefficients (GA and PNA in the right panels) at significant fitting vertices with $p_{FWE} < 0.05$ for each system are illustrated in mean ± SD. All group comparisons are significant ($p_{FDR} < 0.05$, two-sided) and t-values are illustrated by the size of gray circular bubble. The sample size which equals to the number of significant vertices for each bar are listed in Supplementary Table 7). **d** Age information of the preterm and full-term neonate subsets, including PMA, GA, and PNA, are showed as the mean ± SE. GA and PNA show difference between the preterm ($n = 83$) and the age-matched full-term ($n = 83$) with (GA: t = -18.58, $p = 3.4 \times 10^{-42}$, two sided; PNA: t = 14.46, p = 4.5×10^{-31}, two sided). **e** Two-sample T-test indicates that the averaged vertex-level values of individual MFC of full-term group are higher than that of preterm group ($t = 23.42$, $p = 1.8 \times 10^{-119}$, two-sided). Boxes display the interquartile range (IQR: lower hinge: 25th percentile; upper hinge: 75th percentile; center line: median). **f** Spatial maps of averaged vertex-level MFC corresponding to panel e. GA gestational age, PNA postnatal age, PMA postmenstrual age. Source data are provided as a Source Data file. Analyses and visualizations here were performed using MATLAB R2022a (MathWorks Inc., Natick, MA, USA).

circuits, as opposed to merely the maturation of individual white matter tracts[44,52,53]. Building on another recent report that white matter develops more rapidly near the cortical surface[22], our findings could support that the coordinated cortical maturation guides the parallel development of underlying white matter tracts and functional pathways. This also highlights the need to further explore the coupling mechanism of parallel structure/function developments at the connection level. Regarding the two axes, the subject-level myelination covariance (sMC) focuses on understanding how specific regions genetically co-develop within the individual brain, capturing coordination of regional maturation stages, while the group-level myelination covariance (gMC) identifies the broader and common developmental trajectories across neonate cohorts with the effects from both genetic and environmental factors. The gMFC and sMFC exhibited distinct spatial patterns and growth effects (see Figs. 2b and 3b and Supplementary Fig. 6d), indicating that both types of covariances were informative. The subject-level covariance provides insight into personalized brain development, and group-level covariance offers a broader perspective on typical and atypical brain maturation patterns.

The observed spatial patterns of MFC are likely rooted in the specialized sensory and motor demands of neonates (see Fig. 2b). The hierarchical MFC growth during the weeks following birth adheres to a highly hierarchical pattern, where more specialized regions (such as the primary sensory cortex, see Fig. 3b) achieve functional synchronization initially, followed by more complex regions involved in cognitive processes like executive function, attention, and social cognition[3,7]. These regions are pivotal for the development of

https://doi.org/10.1038/s41467-026-70660-4

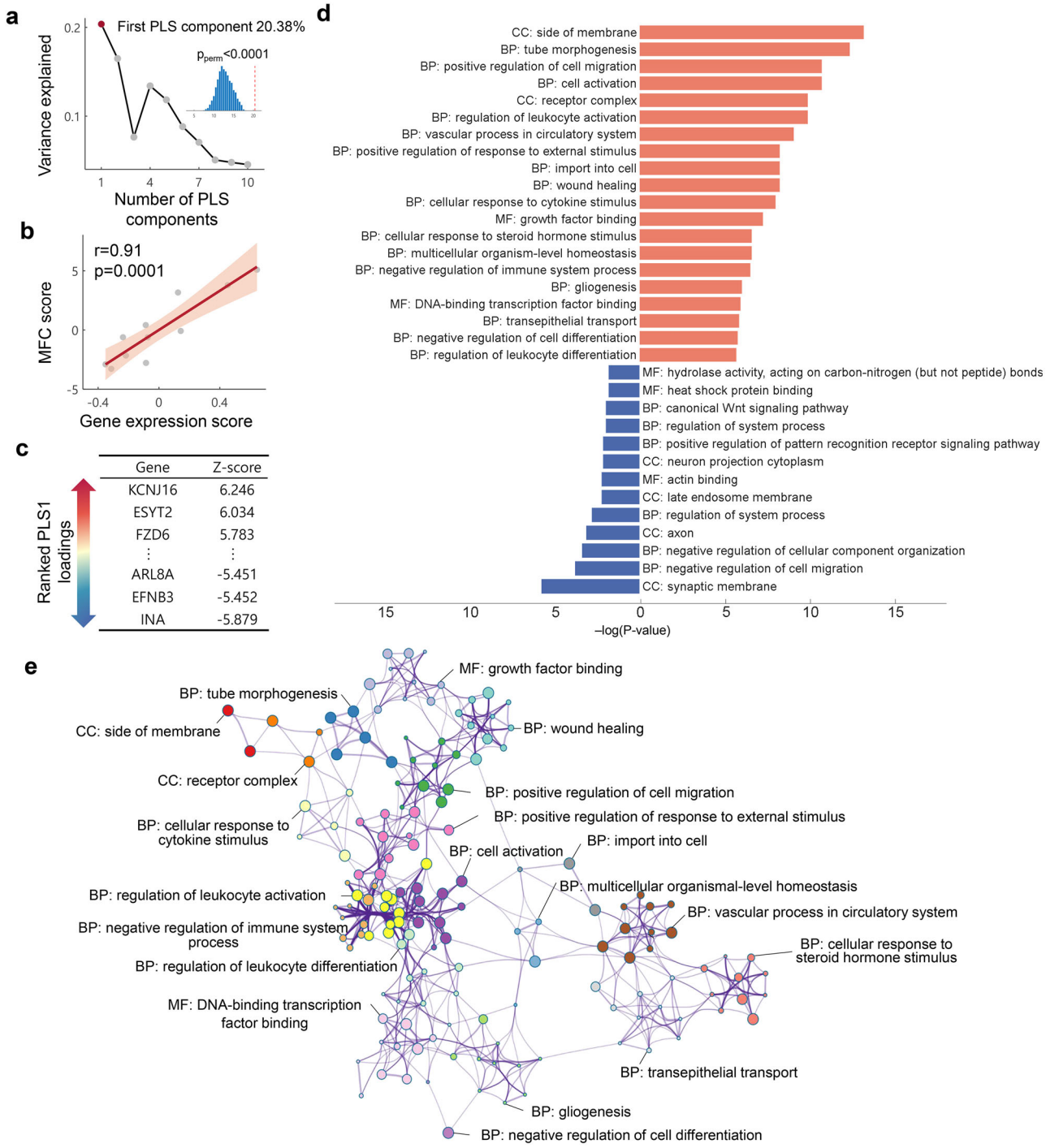

Fig. 6 | Transcriptomic decoding of MFC. a Variance explained for the first 10 components derived from the PLS regression analysis. Permutation test is used by randomly shuffling the orders of regional MFC 5,000 times to assess the statistical significance of PLS1. **b** Pearson's correlation between MFC score and gene expression score ($n = 11$, $r = 0.91$, $p = 0.0001$, CI = [0.90, 0.92], two-sided). Red line and shaded represent the linear regression line with 95% confidence intervals. **c** Ranked PLS1 loadings of significantly associated genes. **d** Enrichment analysis of PLS+ genes (orange bars, Top 20 GO enriched clusters) and PLS- genes (blue bars) associated with MFC, all -log(*P*-value) > 2.22. **e** The intra- and inter-cluster similarities of enriched terms of PLS+ genes are visualized by the Metascape enrichment network. CC Cellular Component, BP Biological Process, MF Molecular Function. Source data are provided as a Source Data file. Analyses and visualization were performed using custom scripts in MATLAB R2022b (MathWorks, Natick, MA, USA).

fundamental sensory and motor functions, which are crucial for survival and interaction with the environment shortly after birth[5,60]. The primary motor cortex and primary sensory cortices, including auditory and visual areas, are among the earliest regions to mature, as they are directly involved in foundational sensory processing and motor coordination. The insula cortex is implicated in a spectrum of functions, plays a vital role in integrating sensory information from the body, and is critical for early bodily awareness and motor planning[61]. The high MFC in these areas reflects both a local maturation process and their homogeneity, which is essential for integrating sensory experiences with motor actions, thereby laying the groundwork for more complex functional integration and behaviors as the child

https://doi.org/10.1038/s41467-026-70660-4

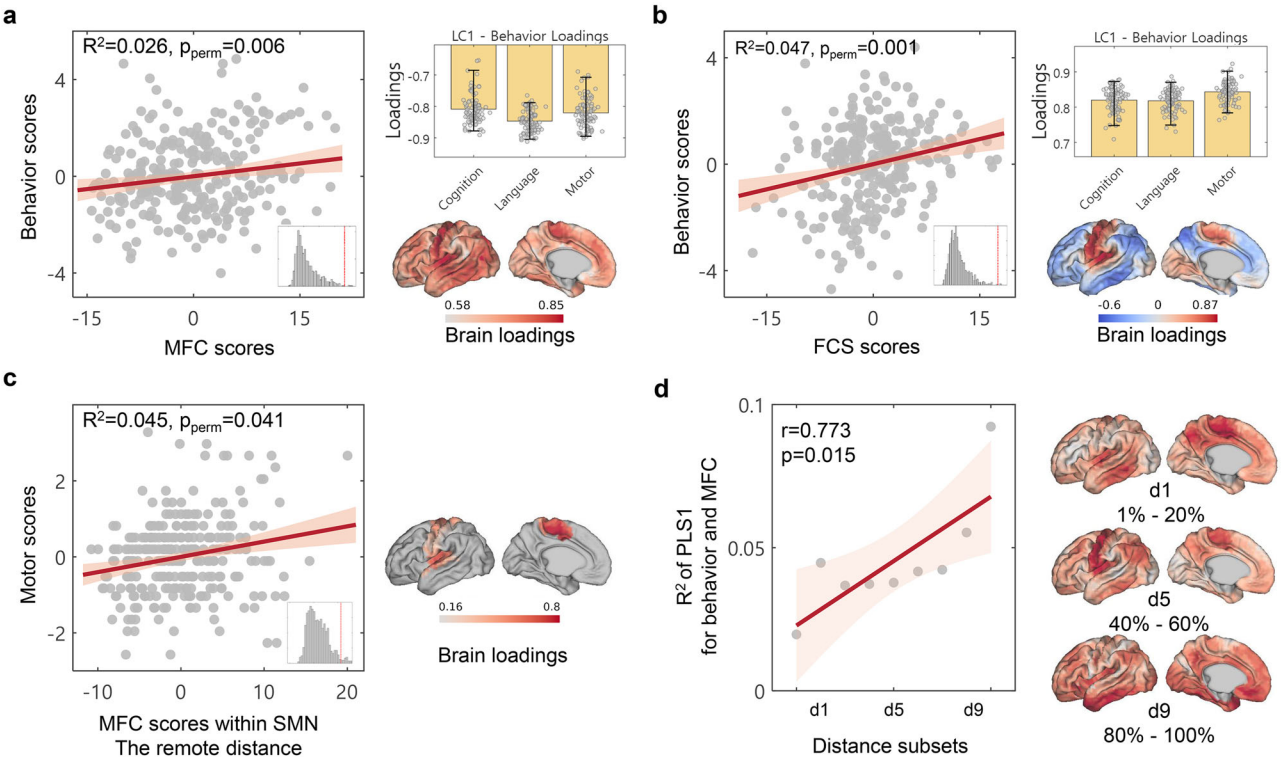

Fig. 7 | Relationship between MFC and the behavior scores at 18 months of age. **a** Significant association between behavioral and MFC composite scores under the PLS1 ($n = 278$, adjusted $R^2 = 0.026$, $p_{perm} = 0.006$, two-sided), with the behavioral and MFC loadings. Red line and shade represent the linear regression line with 95% confidence intervals. Permutation test is used by randomly shuffling the orders of regional MFC 5000 times to assess the association. **b** Significant association between behavioral and FCS composite scores in the first latent component ($n = 278$, adjusted $R^2 = 0.047$, $p_{perm} = 0.001$, two-sided), with the behavioral and FCS loadings. Red line and shade represent the linear regression line with 95% confidence intervals. Permutation test is used by randomly shuffling the orders of regional MFC 5,000 times to assess the association. **c** Significant association between the motor domain and MFC of motor network composite score in the first latent component ($n = 278$, adjusted $R^2 = 0.045$, $p_{perm} = 0.041$, two-sided), with the domain loading and the MFC loadings. Red line and shade represent the linear regression line with 95% confidence intervals. Permutation test is used by randomly shuffling the orders of regional MFC 5000 times to assess the association. **d** The monotonic relationship between the R^2 and the distance ($n = 9$, $r = 0.773$, $p = 0.015$, two-sided), together with the MFC loadings for each distance. Red line and shade represent the linear regression line with 95% confidence intervals. Source data are provided as a Source Data file. Permutation test is used by randomly shuffling the orders of regional MFC 5000 times to assess the association. Analyses and visualization were performed using custom scripts in MATLAB R2022b (MathWorks, Natick, MA, USA) together with Connectome Workbench.

matures[18,34,44]. This functional coherence is not merely a consequence of axonal myelination but is also supported by the temporal and spatial organization of myelination patterns within the cortex (see the similar yet not identical patterns depicted in either the group mean map or the growth rate map between the group-level myelination and the functional connectivity strength in Supplementary Fig. 11).

The development trajectories of MFC across the cortex are not uniform and appear to be shaped by distinct factors before and after birth, as reflected in our dual-axis framework. The results that separating GA and PNA provided a better fit for MFC development than using PMA alone underscore birth as a critical transition. Prenatally, MFC growth was significantly and positively associated with GA across the entire cortex, most prominently in the motor and insula cortex (Fig. 5b, c). This reflects the unfolding of intrinsic genetic programs that orchestrate heterochronic cortical maturation, and this timing may be captured by the subject-level coupling, which also showed dominant MFC in these regions (Figs. 2c and 3c) compared with the group-level. Postnatally, our data revealed a more nuanced picture. While extrauterine development (PNA) also contributed to MFC growth in most regions, its effect size was significantly smaller than that of GA (Fig. 5b, c). Strikingly, the primary motor cortex exhibited a negative MFC growth. This unique postnatal reduction in the motor cortex reflects an activity-dependent refinement or pruning of functional connections. As motor develops prenatally and explodes after birth, the initially broad, genetically specified synchrony may be selectively sharpened, weakening less-relevant connections to enhance the efficiency of frequently used motor circuits[3,18,20,57]. This interpretation is further supported by the finding that full-term neonates (experiencing longer postnatal exposure) had higher MFC in associative regions like the IPL and SFG, but not necessarily in primary motor areas, compared to preterm neonates at term-equivalent age (Fig. 5e, f). Early myelination is not only due to genetic factors but also in response to environments, where sensory and motor experiences drive the neuroplastic growth of these regions[3,62]. Experience-dependent plasticity, along with myelination, especially caused by the birth effect of explosive environmental information, serves as a key driving mechanism. This mechanism links the microstructural aspects, including myelination, with the functional organization or emergence of functional networks during infancy.

The stronger coupling observed in local connections compared with remote connections suggests that early functional integration in the brain occurs primarily within more proximally located cortical regions. This observation aligns with the fact that local regions tend to myelinate synchronously with the genetic influences and lower growth costs[14,57,63], indicating shared developmental timelines and functional requirements. However, the developmental dynamics unveil a more strategic prioritization. While local connections exhibited high baseline MFC, their postnatal growth rate was consistently near zero (Fig. 4b, dashed curves). In stark contrast, long-range connections, despite starting with weaker coupling, demonstrated a significantly

higher and positive growth rate that increased with distance (Fig. 4b, c). This pattern involves balancing the costs against the global efficiency during brain wiring[28,53] in the neonatal brain. The growth of these long-range connections is slower, as these areas need to synchronize their activity over extended distances, which requires additional time to establish the necessary myelination[6–8].

In addition, the transcriptomic associations identified in this study delineate a potential molecular blueprint underlying the observed myelination-function coupling, offering mechanistic hypotheses for how coordinated myelination covaries with the initial assembly of functional pathways. Our gene co-expression analysis revealed that regions with high MFC (PLS1+) are characterized by the upregulation of genes significantly enriched in biological processes crucial for constructing and maintaining the neurovascular coupling and the glial environment. Specifically, enrichments in "transepithelial transport" and "side of membrane" functions suggest the maturation of the blood-brain barrier (BBB)[64,65] and the cell membrane's signaling apparatus, which are vital for maintaining ionic homeostasis and facilitating efficient neurotransmission. Crucially, the significant enrichment of "gliogenesis" terms, particularly those related to astrocyte and oligodendrocyte precursor development, bridges these vascular and neuronal components. Astrocytes are central players in neurovascular coupling[9,14], dynamically regulating local blood flow in response to neuronal activity, while oligodendrocytes are responsible for myelination itself, and microglia modulate learning and neurogenesis[66–68].

In contrast, the genes whose expression was negatively associated with MFC (PLS1-) were enriched for components intrinsic to the post-mitotic neuronal structure and its refinement, such as "synaptic membrane", "neuron projection cytoplasm" and "axon". This pattern may reflect a complementary, and perhaps temporally later, phase of circuit maturation. The relative downregulation of genes involved in axonal growth and synaptic adhesion in high MFC regions could indicate a state of increased circuit stability or a shift towards activity-dependent pruning[20]. In this view, once the foundational neurovascular-glia unit is established via PLS1+ gene activity, securing basic metabolic and signaling efficiency, the focus of development may shift towards the activity-dependent refinement of neuronal connectivity. The lower expression of PLS1- genes might therefore signify a transition from exuberant growth to the selective stabilization of the most functionally relevant synapses and projections within the co-varying regions. This molecular model provides a testable hypothesis for how myelination covariance couples with the sequential stages of functional pathway formation from initial synchronization to refined efficiency.

Another notable validation of our MFC framework lies in its significant association with later behavioral outcomes at about 18 months of age. These associations are not merely global but exhibit a striking functional and anatomical specificity that aligns with the proposed developmental mechanism. At the whole-brain level, MFC showed a comparatively lower association with behavioral outcomes than FCS. This pattern is informative rather than contradictory, as it likely reflects the distinct neurobiological processes captured by each metric and the developmental stage of our cohort. FCS predominantly measures the magnitude of spontaneous neural co-activation, and MFC quantifies the alignment between myelination covariance and this functional co-activation. In the neonatal period, this alignment may be a more nascent and regionally specific process, and only the index explains a significant yet partial portion of the variance in functional connectivity. Functional networks are supported by a multiplex of structural substrates that operate across different spatial and temporal scales. Our study highlights intracortical myelination covariance as a critical, early-developing component of this multiplex, particularly relevant during the neonatal period of profound white matter immaturity. Furthermore, the strength of the MFC-behavior association exhibited a clear distance-dependent gradient, increasing monotonically from the shortest to the longest connections. This observation indicates that long-distance integration, which involves the synchronization of distant brain regions, plays a crucial role in generating behaviorally relevant connections[9,69]. Long-distance coupling supports the notion that the integration of more distant cortical areas contributes to the development of complex cognitive functions and behaviors. These findings emphasize that long-range connections, despite their relatively later myelination, are integral to shaping the brain's ability to coordinate complex behaviors[8,55,69].

While this study provides valuable insights into early brain development, several limitations must be acknowledged. First, the study's focus on gray matter myelination may overlook some of the finer details of white matter development, as the MRI techniques used are more sensitive to cortical gray matter myelination than the ongoing maturation of white matter in the neonate brain. Including various information, either the microstructure or bold signals of the white matter[59,70], to support the brain functional connectome is beneficial to give a complete picture of how early brain function emerges. The complexity of structural covariance networks, influenced by genetic, epigenetic, and environmental factors, was not fully addressed in this study, and future research incorporating these factors is needed. Second, the T1w/T2w ratio was used to quantify myelin content in this study. Given its ease of calculation from commonly acquired images, T1w/T2w is usually used in large-scale datasets and/or early life samples where acquisition times are particularly restricted. It must be admitted that the validity of T1w/T2w as a marker of myelin remains debated[71,72], although studies comparing T1w/T2w to R1 have validated its usability in adult cortex[73] and infant white matter[74]. Third, the cross-sectional design restricts our ability to sufficiently capture dynamic changes in myelination and functional connectivity over time within a subject, emphasizing the need for longitudinal studies to track developmental trajectories for a wide range of ages. Fourth, the resolution and sensitivity of MRI in neonates may limit the ability to detect small-scale myelin changes as well as other microstructures along the gradient across cortical layers, especially in the early stages[14,24]. With high-resolution data, the subject-level myelination covariance can be replaced by any other structure similarity along the depth gradient of cortical layers[53]. Finally, while associations between myelination patterns and functional connectivity are demonstrated, the causal mechanisms linking myelination covariance to functional network emergence remain unclear, suggesting that other neurodevelopmental processes may also contribute. Experimental studies, including animal models, will be crucial to better understand the causality of these relationships.

Methods

Participants and data acquisition

A total of 887 scan sessions from 783 neonatal subjects were collected from the Developing Human Connectome project (dHCP). Based on the dataset screened in Logan's research[75], we further excluded subjects with missing myelin maps, calculated with the T1w/T2w ratio. Finally, 447 scans (447 subjects) were included in the following analysis, in which 364 were termborn (female/male = 167/197; GA: 39.65 ± 1.32 weeks; PMA at scan: 40.80 ± 1.96 weeks), and 83 were preterm-born (female/male = 34/49; GA: 32.08 ± 3.47 weeks; PMA at scan: 40.73 ± 2.09 weeks). The dHCP study was approved by the UK Health Research Authority (14/LO/1169), and written consent was obtained from the parents or legal guardians of the study subjects. All neuroimaging data were acquired at the Evelina Newborn Imaging Centre, Evelina London Children's Hospital, using a 3-Tesla Philips Achieva system. Detailed information on data acquisition parameters is available on the dHCP website (http://www.developingconnectome.org/).

MRI data preprocessing

All anatomical and functional MRI data were minimally preprocessed with the details of the dHCP structural pipeline[76], and the functional pipeline[77]. Briefly, the surface-based pipeline for structural images included superresolution reconstruction to obtain the 3D T1w/T2w volumes[78], registration (from T1w to T2w), bias correction, brain extraction, segmentation[79], surface extraction[80], and surface registration[81]. Furthermore, for the anatomical data, T1w/T2w intensity values were sampled at the mid-thickness surface. We used the new version of the Multimodal Surface Matching (newMSM[81,82]) to perform a non-linear alignment of surfaces and T1w/T2w values of each neonate to the dHCP age-specific templates, which reside in the HCP-YA fs_LR 32k space and cover the age range from 28 to 40 weeks PMA[75,83]. Subsequently, data registered to age-specific templates were aligned to the 40-week PMA template, downsampled to a 5k_fs_LR mesh, and lightly smoothed with 1.5 mm full-width-halfmaximum (FWHM) using workbench[84] to better simulate the spatial continuity.

For the rs-fMRI data, the minimally pre-processed 4D functional volume images, which were motion and distortion corrected (MCDC) with FSL EDDY, denoised based on spatial independent component analysis (sICA) and nuisance regression, temporally band-pass filtered (0.01–0.09 Hz) and spatial registration. Even though the potential head motion effects on fMRI signals had been carefully controlled in MCDC, sICA and motion parameters regression, we further computed the relationship between mean relative root mean square (RMS) displacement and functional connectivity (QC-FC correlations)[85], resulting that the distribution of QC-FC correlations was zero-centered with no significance. The residual mean framewise displacements (mFDs) were further examined for age effects, and nothing significant was found. For the registration challenge since the fast development of infant brain, a two-step template alignment method was adopted, with a non-linear registration from native functional space to each age specific template[80] (dHCP volume atlas, https://gin.g-node.org/BioMedIA/dhcp-volumetric-atlas-groupwise), which included templates for 36 to 44 weeks of postmenstrual age, and another linear alignment from each age specific template to the target 40-week PMA template (dHCP age specific template). The two transform files were estimated and applied to the functional volumes using Advanced Normalization Tools (ANTS, http://stnava.github.io/ANTs/). Finally, the cleaned rs-fMRI data were further projected onto the individual native cortical surface and then spatially registered and smoothed using surface pipelines.

Functional connectivity network

Functional connectivity (FC) between each pair of vertices was quantified as the Fisher-Z-transformed Pearson correlation between the pairwise BOLD time series. For each participant, an 8589 × 8589 (8589 is the total number of vertices in the left and right hemispheres after removing subcortical structures at the 5k surface resolution) weighted adjacency matrix encoded the functional connectivity network. In addition, another two FC matrices with different cortical granularities were generated for the Icosahedron-1002 atlas[86,87] and the neonatal AAL atlas[88]. Removing the subcortical regions, the neonatal AAL atlas included 82 cortical regions, and the Icosahedron-1002 atlas included 923 uniform cortical regions within each hemisphere.

White matter-based structural connectivity network

Structural connectivity (SC) between each pair of regions was calculated as the fiber tract connectivity density using DTI imaging data for fiber tract reconstruction based on the ROI-level atlas. We utilized two alternative volume-based infant atlases: the UNC infant AAL atlas[88], comprising 8 subcortical structures and 82 cortical structures. The reconstruction of the diffusion data was performed in native space with DSI-studio (https://dsi-studio.labsolver.org/). After the generalized q-sampling imaging (GQI) with a diffusion sampling length ratio of 1.25, the whole-brain fiber tracking was conducted with the parameters including the angular cutoff of 60 degrees, step size of 1.0 mm, minimum length of 30 mm, and maximum length of 300 mm. The whole-brain fiber tracking process was performed with the FACT algorithm until 1,000,000 streamlines were reconstructed for each individual. The fiber tract connectivity density was quantified as the number of streamlines normalized by volumes of paired regions, yielding a 90 × 90 (90 parcellations with 82 cortical regions and 8 subcortical regions) matrix encoded the structural connectivity network (SCN) for each participant. Then, Sparse structural connections were converted into a communicability matrix to avoid the loss of structural information, resulting in a fully weighted form of the SCN based on the communication model[89,90].

Dual-axis myelination covariant network

In this study, we proposed a dual-axis myelination covariance framework to quantify the spatiotemporal homogeneity of myelin developmental dynamics during infancy. The vertex-wise myelin (T1w/T2w) was first Z-normalized to quantify myelination levels relative to whole-brain baselines within each individual. On this basis, the dual-axis MC was derived: (1) group-level myelination covariance (gMC) across subjects captures inter-subject covariance, which is defined as the Fisher-Z-transformed Pearson correlation between each pair of cortical vertices using Z-normalized myelin data connected as an N × L sequence across subjects (where N is the number of subjects, 364 for term-born and 83 for preterm-born; L is the number of vertices), yielding an 8589 × 8589 covariance matrix. (2) Subject-level myelination covariance (sMC) captures individual-level covariance patterns, defined as the intra-subject dot products of the Z-normalized myelin values, independent of confounding extraneous subject-level variables, yielding another 8589 × 8589 covariance matrix.

Calculation of myelination-function coupling

Vertex-wise connectivity profiles, represented as vectors of connectivity strength of a vertex to all other vertices in the whole brain, were extracted from each column of the myelination covariance connectome or functional connectome. Myelination-function coupling (MFC) was then measured using a linear regression model. For a given vertex, the predictive variable was the myelination covariance (MC) profile, and the functional connectivity (FC) profile was the dependent variable. As shown in the univariate model:

$$FC = b_0 + b_1 \times MC, \tag{1}$$

where the intercept b_0 and regression coefficients b_1 were estimated model parameters, MC was gMC profile or sMC profile corresponding to the calculation process of gMFC or sMFC. In the present study, we assumed that a linear model with a dual-axis myelination covariance matrix as multiple predictors captures more comprehensive and integrated information about myelination-function coupling. As shown in the bivariate model:

$$FC = b_0 + b_1 \times gMC + b_2 \times sMC. \tag{2}$$

The goodness of fit per vertex represents the vertex-level MFC, quantified as the adjusted coefficient of determination (R^2_{adj}). The R^2_{adj} from the univariate model was used in the quantification of gMFC or sMFC, and the R^2_{adj} from the bivariate model was used to quantify MFC. Vertex-wise indices of MFC were averaged across participants to create a mean vertex-level coupling map (Fig. 2c).

In general, the coefficient of determination (R^2) tends to increase when the number of parameters included in the model increases. One strategy to account for this addition is to adjust the R^2 measure based on the number of predictors. Specifically, we calculated R^2_{adj} for each

brain region or vertex:

$$R^2_{adj} = 1 - \frac{N-1}{N-p-1} \cdot \left(1 - R^2\right), \quad (3)$$

where N is the number of samples and p is the number of predictors in the model, and is equal to p=2 for the dual-axis framework (bivariate model, including both sMC and gMC).

Age effects analysis of MFC

To further validate the development of cortical MFC in human infants, we considered age as a continuous variable using a generalized additive model (GAM)[91] to flexibly investigate linear and nonlinear relationships between MFC and age. For each cortical vertex, the GAM model was defined as follows:

$$MFC_{i,subj} = \beta_{i,0} + \beta_{i,1} \times f\left(PMA_{subj}\right) + covs, \quad (4)$$

where the MFC of participant *subj* was the dependent variable, *i* represented the vertex, $f(.)$ was a smooth term and PMA_{subj} was the age of *subj* at scan. Importantly, thin plate regression splines were used for the smoothing basis, and the residual estimates of the maximum likelihood (REML) method was used to estimate nonlinearities, penalizing nonlinearity to avoid over-fitting the data[92]. The averaged first derivative of the age smooth function (ΔMFC/Δ PMA) characterized the age effects on MFC development. We controlled false positive errors for multiple comparisons using the False Discovery Rate ($p_{FDR} < 0.05$). The *covs* was co-variables including head size, sex and mean framewise displacement (mFD). In addition, linear age effects were also investigated using a linear regression model. Significance inference of linear effects was performed using surface-based, vertex-wise permutation testing ($p_{FWE} < 0.05$; FSL PALM[93], version alpha119) with threshold-free cluster enhancement (TFCE)[94].

Distance dependence analysis of MFC

Considering that spatially adjacent vertices may intrinsically overestimate the coupling between myelination and functional profile, the three types of connection profiles (FC, gMC, sMC) were ranked according to the geodesic distance between vertices and equally separated into multiple subsets. This cortical geodesic distance between all pairwise vertices was calculated according to the workbench command "surface-geodesic-distance-all-to-all" on the mid-thickness surface, with values ranging from 0.75 mm to 139.11 mm (the mean distance for node subsets is detailed in Supplementary Table 8). Therefore, a series of MFCs was calculated to assess the coupling for the connections between nodes with various cortical distances, yielding a distance-specific examination for MFC. Detailed nine subsets (about 1/9 = 11% for each, mean value 18.47 mm for the most local) and fifty subsets (1/50 = 2% for each, mean value 8.12 mm for the most local) of the distances were selected to show distance-specific MFC patterns as well as the variation curves for systems or core regions. Furthermore, for each distance-specific MFC, the growth rates were assessed with PMA for each node to identify the growth sensitivity of distance-specific MFC. From the perspective of brain system integration, we further examined the distance-dependent properties of MFC within seven functional networks and five cytoarchitectonic systems. The MFC and its age effects for each system or network were derived by averaging across all constituent nodes.

Birth event effect analysis of MFC

To further explore the birth effects, we utilized a comparison between an early, experience-independent phase and a subsequent experience-dependent phase. Two distinct phases were separately represented by gestational age (GA) and postnatal age (PNA). PNA was the difference between GA at birth and the postmenstrual age (PMA) at scan. Therefore, two linear models were fitted at the vertex level for the MFC, with Model I for the PMA effect:

$$MFC_{i,subj} = \beta_{i,0} + \beta_{i,1} \times PMA_{subj} + covs, \quad (5)$$

and Model II for GA and PNA effects, accounting for the development of extrauterine or intrauterine,

$$MFC_{i,subj} = \beta_{i,0} + \beta_{i,1} \times GA_{subj} + \beta_{i,1} \times PNA_{subj} + covs, \quad (6)$$

where the GA_{subj} and PNA_{subj} were the age at birth and the age at scan after birth. The birth effect was indicated by the difference between Model I and Model II, considering the adjusted R^2 of each model and the linear coefficients of the age terms. To disentangle the influence of early postnatal experience from endogenous developmental processes, we compared the group mean MFC in two sub-cohorts matched on PMA: a preterm-born cohort (N = 83; female/male = 34/49; GA: 32.08 ± 3.47 weeks; PMA at scan: 40.73 ± 2.09 weeks) and a matched term-born cohort (N = 83; female/male = 38/45; GA: 39.65 ± 1.32 weeks; PMA at scan: 40.80 ± 1.96 weeks). As the preterm-born cohort had a shorter gestational age but longer postnatal experience at the time of scan, any differences in MFC patterns can be more confidently attributed to the effects of birth and ex-uterine experience. Significance inference of the linear modeling analysis was conducted with permutation testing (FSL PALM[93]) with TFCE[94] under $p_{FWE} < 0.05$.

Spatial association with transcriptional gene expression

To elucidate the underlying biological mechanisms of myelination-function coupling during neonates, this study utilized transcriptomic data from donors aged 24 post-conceptual weeks (pcw) and 37 pcw obtained from the BrainSpan Atlas (https://www.brainspan.org/static/home) of the Developing Human Brain. Samples with insufficient regional coverage were excluded under the assumption that the BrainSpan cohort and our study population share homogeneity. RNA-Seq expression averaged values were extracted from 11 brain regions, including the primary sensorimotor cortices (M1C, V1C, A1C, S1C) and other higher-order cortices (see the Supplementary Table 9 for details). Then, PLS (partial least squares) regression was employed to probe the potential relationship between MFC and transcriptional levels for all genes. PLS1(first component of PLS) depicted the most correlated cortical expression map with respect to the MFC. Additionally, we performed a nonparametric permutation test by randomly shuffling the orders of regional MFC 5000 times to assess the statistical significance of PLS1. The bootstrapping technique was performed with 5000 bootstrap replicates to evaluate the contribution of each gene to the PLS1. The Z-score per gene was determined by dividing its weight by the corresponding bootstrap standard error, based on which all genes were ranked. Then, genes exhibiting $Z > 3$ and $p_{FDR} < 0.001$ were assigned to the PLS1+ gene list, while those with $Z < -3$ and $p_{FDR} < 0.001$ were allocated to the PLS1- gene list. Subsequently, the PLS1+ and PLS1- gene lists were separately subjected to the Metascape (https://metascape.org/gp-/index.html) to identify enriched GO terms, including biological process (BP), molecular function (MF), and cellular component (CC).

Analysis of behavioral outcomes

To investigate associations between the proposed MFC and the neurodevelopmental outcomes, we analyzed the cortical measures from the neonatal data together with their follow-up behavior assessment at about 18 months of age. Vertex-wise MFC maps were parcellated into 923 uniform cortical regions for each hemisphere to generate the individual MFC feature vector, yielding a 1 × 923 MFC feature vector (averaging from the two hemispheres). To compare, another individual cortical measure, the functional connectivity strength (FCS), also

extracted into a 1 × 923 FCS feature vector and linked with the behavioral outcomes. Neurodevelopmental outcomes were assessed using the Bayley Scales of Infant Development III, including[95,96] composite scores for: motor development (fine/gross motor skills), language development (receptive/expressive communication), and cognitive development (sensory perception, problem-solving). PLS regression[95,96] was applied to the brain matrix (participants × 923 cortical measures) and behavior composite scores (participants × 3 Bayley domains). Significance of PLS1 was determined through 5000 times permutation test ($p_{perm} < 0.05$) for multiple comparisons. To interpret the PLS1, we computed PLS loadings, measured by Pearson's correlations between the original brain data and brain composite scores, as well as between the original behavior data and behavioral composite scores for PLS1. To estimate confidence intervals for these loadings, we applied a bootstrapping procedure that generated 500 samples from subjects' brain and behavior data. Code for this analysis can be found here: https://github.com/MIPLabCH/myPLS.

Null model

We tested the observed coupling against the spatial permutation test ("spin test"). Using this null model, we explored whether the observed coupling was specific to the actual connectivity pattern rather than due to the spatial autocorrelation. Specifically, we first recorded the spherical coordinates of each vertex at a hemispherical resolution of 5k. Then, we randomly rotated the vertices while maintaining spatial autocorrelation and reassigned vertex values to the nearest vertices. This procedure was repeated 5000 times. We then calculated the MFC of null model to build a distribution of null maps that preserve spatial neighborhood information. The spin-based p-value was calculated as the proportion of MFC in the null model that exceeded (for positive coupling) or were weaker than (for negative coupling) the empirical MFC.

Connectome gradients

To identify spatial axes in connectivity variation across different areas, we applied an unsupervised technique called diffusion map embedding to compute gradients of the three connectomes separately (mFC, gMC, sMC). This technique converted the connectivity matrix into a normalized angle matrix, which scales the angle between each pair of vertices as a function of similarity. Manifold learning parameters were identical to those previously described, specifically $\alpha = 0.05$ and automated diffusion time estimation[97]. The computation was done with the MATLAB BrainSpace package (https://www.mathworks.com/matlabcentral/fileexch-ange/86887-brainspace).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The MRI data used in this study are available in the dHCP database and can be downloaded through the website at https://www.developingconnectome.org/. The BrainSpan Atlas dataset of the developing human brain is publicly available at https://www.brainspan.org/static/home. DhcpSym spatiotemporal cortical surface atlas at https://brain-development.org/brain-atlases/atlases-from-the-dhcp-project/cortical-surface-template/; The standard human cortical surface-based Icosahedron-1002 atlas used in this study is publicly available at https://github.com/DiedrichsenLab/fs_LR_32; The volume-based UNC infant AAL atlas is publicly available at https://www.nitrc.org/projects/pediatricatlas; The seven cortical networks parcellation on surface by Thomas Yeo at https://surfer.nmr.mgh.har-vard.edu/fswiki/CorticalParcellation_Yeo2011; Free-Surfer *fsaverage* space: https://github.com/WashingtonUniversity/HCPpipelines/tree/master/global/te-mplates/standard_mesh_atlases/resample_fsaverage; Demographic data for the dHCP are available at https://github.com/BioMedIA/dHCP-release-notes/blob/master/supplementary_files/combined.tsv. Source data are provided with this paper.

Code availability

The dHCP-structural-pipeline at https://github.com/BioMedIA/dhcp-structural-pipeline; dHCP-functional-pipeline at https://git.fmrib.ox.ac.uk/seanf/dhcp-neonatal-fmri-pipeline; Connectome Workbench https://www.humanconnecto-me.org/software/connectome-workbench; dHCP MSM configuration file: https://github.com/ecr05/dHCP_template_alignment/blob/master/configs/config_subject_to_40_week_template_3rd_release; FSL PALM: https://github.com/andersonwinkler/PALM; Advanced Normalization Tools (ANTS) at http://stnava.github.io/ANTs/; The codes of PLS analysis used in brain behavior association are avilabel at: https://github.com/MIPLabCH/myPLS; the MATLAB BrainSpace package for connectome gradient at https://www.mathworks.com/matlabcentral/fileexch-ange/86887-brainspace; Gene Ontology(GO) enrichment analysis using Metascape at https://metascape.org/gp/index.html; The custom MATLAB (R2022a) codes for the proposed MFC calculation as well as the other organized codes are publicly available from the GitHub at https://github.com/NI-Connectome/Myelination-function-coupling and archived at Zenodo (https://doi.org/10.5281/zenodo.18595951).

References

1. Turk, E. et al. Functional connectome of the fetal brain. *J. Neurosci.* **39**, 9716–9724 (2019).
2. Keunen, K., Counsell, S. J. & Benders, M. J. N. L. The emergence of functional architecture during early brain development. *Neuroimage*. **160**, 2–14 (2017).
3. Gao, W. A hierarchical model of early brain functional network development. *Trends Cogn. Sci*. **29**, 855–868 (2025).
4. Gao, W. et al. Evidence on the emergence of the brain's default network from 2-week-old to 2-year-old healthy pediatric subjects. *Proc. Natl. Acad. Sci. USA*. **106**, 6790–6795 (2009).
5. Herschkowitz, N. Brain development in the fetus, neonate and infant. *Neonatology* **54**, 1–19 (1988).
6. Gao, W., Alcauter, S., Smith, J. K., Gilmore John, H. & Lin, W. Development of human brain cortical network architecture during infancy. *Brain Struct. Funct*. 1173–1186 https://doi.org/10.1007/s00429-014-0710-3 (2014).
7. Gilmore, J. H., Knickmeyer, R. C. & Gao, W. Imaging structural and functional brain development in early childhood. *Nat. Rev. Neurosci.* **19**, 123–137 (2018).
8. Cao, M., Huang, H. & He, Y. Developmental connectomics from infancy through early childhood. *Trends Neurosci.* **40**, 494 (2017).
9. Fotiadis, P. et al. Structure–function coupling in macroscale human brain networks. *Nat. Rev. Neurosci*. **25**, 688–704 (2024).
10. Tooley, U. A. et al. Structure–function coupling in the first month of life: Associations with age and attention. *Proc. Natl. Acad. Sci. USA*. **122**, e2412729122 (2025).
11. Hutchison, R. M. & Morton, J. B. Tracking the brain's functional coupling dynamics over development. *J. Neurosci.* **35**, 6849–6859 (2015).
12. Wang, R. et al. The overgrowth of structure-function coupling in the premature brain during infancy. *Dev. Cogn. Neurosci.* **72**, 101535 (2025).
13. Feng, G. et al. Spatial and temporal pattern of structure–function coupling of human brain connectome with development. *Elife*. **13**, RP93325 (2024).
14. Baum, G. L. et al. Development of structure–function coupling in human brain networks during youth. *Proc. Natl. Acad. Sci. USA*. **117**, 771–778 (2020).

15. Cheema, K. & Cummine, J. The relationship between white matter and reading acquisition, refinement and maintenance. *Dev. Neurosci.* **40**, 209–222 (2018).
16. Preti, M. G. & Van De Ville, D. Decoupling of brain function from structure reveals regional behavioral specialization in humans. *Nat. Commun.* **10**, 4747 (2019).
17. Soman, S. M. et al. Cortical structural and functional coupling during development and implications for attention deficit hyperactivity disorder. *Transl. Psychiatry* **13**, 252 (2023).
18. Natu, V. S. et al. Infants' cortex undergoes microstructural growth coupled with myelination during development. *Commun. Biol.* **4**, 1–12 (2021).
19. Bonetto, G., Belin, D. & Káradóttir, R. T. Myelin: a gatekeeper of activity-dependent circuit plasticity? *Science*. **374**, eaba6905 (2021).
20. Faust, T. E., Gunner, G. & Schafer, D. P. Mechanisms governing activity-dependent synaptic pruning in the developing mammalian CNS. *Nat. Rev. Neurosci.* **22**, 657–673 (2021).
21. Zika, S. et al. Cortical and white matter myelination proceed in concert during early infancy. Preprint at https://doi.org/10.1101/2025.07.07.663449 (2025).
22. Pérez, K. et al. Hierarchical microstructural tissue growth of the gray and white matter of human visual cortex during the first year of life. *Brain Struct. Funct.* **231**, 15 (2026).
23. Bobba, P. S. et al. Age-related topographic map of magnetic resonance diffusion metrics in neonatal brains. *Hum. Brain Mapp.* **43**, 4326–4334 (2022).
24. Grotheer, M. et al. White matter myelination during early infancy is linked to spatial gradients and myelin content at birth. *Nat. Commun.* **13**, 997 (2022).
25. Tau, G. Z. & Peterson, B. S. Normal development of brain circuits. *Neuropsychopharmacology*. **35**, 147–168 (2010).
26. Sun, L. et al. Human lifespan changes in the brain's functional connectome. *Nat. Neurosci.* **28**, 891–901 (2025).
27. Vianna-Barbosa, R. et al. Myelination of callosal axons is hampered by early and late forelimb amputation in rats. *Cereb. Cortex Commun.* **2**, tgaa090 (2020).
28. Keunen, K. et al. Early human brain development: insights into macroscale connectome wiring. *Pediatr. Res.* **84**, 829–836 (2018).
29. Nozais, V. et al. Atlasing white matter and grey matter joint contributions to resting-state networks in the human brain. *Commun. Biol.* **6**, 726 (2023).
30. Bijsterbosch, J. D., Valk, S. L., Wang, D. & Glasser, M. F. Recent developments in representations of the connectome. *Neuroimage*. **243**, 118533 (2021).
31. Thion, M. S., Ginhoux, F. & Garel, S. Microglia and early brain development: an intimate journey. *Science*. **362**, 185–189 (2018).
32. Thornton, M. A. et al. Long-term in vivo three-photon imaging reveals region-specific differences in healthy and regenerative oligodendrogenesis. *Nat. Neurosci.* **27**, 846–861 (2024).
33. Beul, S. F., Goulas, A. & Hilgetag, C. C. Comprehensive computational modelling of the development of mammalian cortical connectivity underlying an architectonic type principle. *PLoS Comput. Biol.* **14**, e1006550 (2018).
34. Goulas, A., Betzel, R. F., & Hilgetag, C. C. Spatiotemporal ontogeny of brain wiring. *Sci. Adv.* **5**, eaav9694 (2019).
35. Barbas, H. & Hilgetag, C. C. From circuit principles to human psychiatric disorders. *Biol. Psychiatry*. **93**, 388–390 (2023).
36. Burt, J. B. et al. Hierarchy of transcriptomic specialization across human cortex captured by structural neuroimaging topography. *Nat. Neurosci.* **21**, 1251 (2018).
37. Alexander-Bloch, A., Giedd, J. N. & Bullmore, E. Imaging structural covariance between human brain regions. *Nat. Rev. Neurosci.* **14**, 322–336 (2013).
38. Seidlitz, J. et al. Morphometric similarity networks detect microscale cortical organization and predict inter-individual cognitive variation. *Neuron*. **97**, 231–247.e7 (2018).
39. Sebenius, I. et al. Robust estimation of cortical similarity networks from brain MRI. *Nat. Neurosci.* **26**, 1461–1471 (2023).
40. Paquola, C. et al. Shifts in myeloarchitecture characterise adolescent development of cortical gradients. *Elife*. **8**, e50482 (2019).
41. Wu, X. et al. Morphometric dissimilarity between cortical and subcortical areas underlies cognitive function and psychiatric symptomatology: a preadolescent study from ABCD. *Mol. Psychiatry*. **28**, 1146–1158 (2023).
42. Jiang, L. et al. Toward neurobiological characterization of functional homogeneity in the human cortex: regional variation, morphological association and functional covariance network organization. *Brain Struct. Funct.* **220**, 2485–2507 (2014).
43. Huang, Z., Gao, W., Wu, Z., Li, G. & Nie, J. Functional brain activity is highly associated with cortical myelination in neonates. *Cereb. Cortex* **33**, 3985–3995 (2023).
44. Fotiadis, P. et al. Myelination and excitation-inhibition balance synergistically shape structure-function coupling across the human cortex. *Nat. Commun.* **14**, 1–21 (2023).
45. Deoni, S. C. L., Dean, D. C., Remer, J., Dirks, H. & O'Muircheartaigh, J. Cortical maturation and myelination in healthy toddlers and young children. *Neuroimage*. **115**, 147–161 (2015).
46. Glasser, M. F., Goyal, M. S., Preuss, T. M., Raichle, M. E. & Van Essen, D. C. Trends and properties of human cerebral cortex: correlations with cortical myelin content. *Neuroimage*. **93**, 165–175 (2014).
47. Corrigan, N. M. et al. Myelin development in cerebral gray and white matter during adolescence and late childhood. *Neuroimage* **227**, 117678 (2021).
48. Gao, W. et al. Temporal and spatial development of axonal maturation and myelination of white matter in the developing brain. *Am. J. Neuroradiol.* **30**, 290–296 (2009).
49. Gong, G., He, Y., Chen, Z. J. & Evans, A. C. Convergence and divergence of thickness correlations with diffusion connections across the human cerebral cortex. *Neuroimage*. **59**, 1239–1248 (2012).
50. Edwards, A. D. et al. The developing human connectome project neonatal data release. *Front. Neurosci.* **16**, 886772 (2022).
51. Yeo, B. T. T. et al. The organization of the human cerebral cortex estimated by intrinsic functional connectivity. *J. Neurophysiol.* **106**, 1125–1165 (2011).
52. Tijms, B. M., Seris, P., Willshaw, D. J. & Lawrie, S. M. Similarity-based extraction of individual networks from gray matter MRI scans. *Cereb. Cortex*. **22**, 1530–1541 (2012).
53. Sebenius, I. et al. Structural MRI of brain similarity networks. *Nat. Rev. Neurosci.* **26**, 42–59 (2025).
54. Zhou, Y. et al. Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. *Nat. Commun.* **10**, 1523 (2019).
55. Zhu, J. & Yu, Y. Measuring brain structure-function coupling: a promising approach to decode psychiatric neuropathology. *Biol. Psychiatry*. **97**, 212–214 (2025).
56. Bijsterbosch, J. D. et al. The relationship between spatial configuration and functional connectivity of brain regions. *Elife*. **7**, e32992 (2018).
57. Gondová, A., Neumane, S., Arichi, T. & Dubois, J. Early development and co-evolution of microstructural and functional brain connectomes: a multi-modal MRI study in preterm and full-term infants. *Hum. Brain Mapp.* **46**, e70186 (2025).
58. Mohapatra, A. N., Peles, D., Netser, S. & Wagner, S. Synchronized LFP rhythmicity in the social brain reflects the context of social encounters. *Commun. Biol.* **7**, 1–18 (2024).
59. Li, M. et al. Functional contrast across the gray-white matter boundary. *Nat. Commun.* **16**, 1–14 (2025).

60. Hodel, A. S. Rapid infant prefrontal cortex development and sensitivity to early environmental experience. *Dev. Rev.* **48**, 113 (2018).
61. Chen, W. G. et al. The emerging science of interoception: sensing, integrating, interpreting, and regulating signals within the self. *Trends Neurosci.* **44**, 3–16 (2021).
62. Kolb, B. & Gibb, R. Brain plasticity and behaviour in the developing brain. *J. Can. Acad. Child Adolesc. Psychiatry*. **20**, 265 (2011).
63. Saberi, A. et al. The regional variation of laminar thickness in the human isocortex is related to cortical hierarchy and interregional connectivity. *PLoS Biol.* **21**, e3002365 (2023).
64. Gong, Y. et al. Research developments in the neurovascular unit and the blood-brain barrier (Review). *Biomed. Rep.* **22**, 88 (2025).
65. Kaplan, L., Chow, B. W. & Gu, C. Neuronal regulation of the blood–brain barrier and neurovascular coupling. *Nat. Rev. Neurosci.* **21**, 416 (2020).
66. Romero, D. M., Bahi-Buisson, N. & Francis, F. Genetics and mechanisms leading to human cortical malformations. *Semin. Cell Dev. Biol.* **76**, 33–75 (2018).
67. Gelfo, F., Mandolesi, L., Serra, L., Sorrentino, G. & Caltagirone, C. The neuroprotective effects of experience on cognitive functions: evidence from animal studies on the neurobiological bases of brain reserve. *Neuroscience*. **370**, 218–235 (2018).
68. Vieira, M. S. et al. Decoding epigenetic cell signaling in neuronal differentiation. *Semin. Cell Dev. Biol.* **95**, 12–24 (2019).
69. Betzel, R. F. & Bassett, D. S. Specificity and robustness of long-distance connections in weighted, interareal connectomes. *Proc. Natl. Acad. Sci. USA*. **115**, E4880–E4889 (2018).
70. Gore, J. C. et al. The nature and interpretation of BOLD signals in white matter—a review. *Magn. Reson. Imaging*. **127**, 110596 (2026).
71. Uddin, M. N., Figley, T. D., Solar, K. G., Shatil, A. S., & Figley, C. R. Comparisons between multi-component myelin water fraction, T1w/T2w ratio, and diffusion tensor imaging measures in healthy human brain structures. *Sci. Rep.* **9**, 2500 (2019).
72. Sandrone, S. et al. Mapping myelin in white matter with T1-weighted/T2-weighted maps: discrepancy with histology and other myelin MRI measures. *Brain Struct. Funct.* **228**, 525–535 (2023).
73. Shams, Z., Norris, D. G. & Marques, J. P. A comparison of in vivo MRI-based cortical myelin mapping using T1w/T2w and R1 mapping at 3T. *PLoS One*. **14**, e0218089 (2019).
74. Grotheer, M. et al. Human white matter myelinates faster in utero than ex utero. *Proc. Natl. Acad. Sci. USA*. **120**, e2303491120 (2023).
75. Williams, L. Z. J. et al. Structural and functional asymmetry of the neonatal cerebral cortex. *Nat. Hum. Behav*. **7**, 942–955 (2023).
76. Makropoulos, A. et al. The developing human connectome project: a minimal processing pipeline for neonatal cortical surface reconstruction. *Neuroimage*. **173**, 88–112 (2018).
77. Fitzgibbon, S. P. et al. The developing Human Connectome Project (dHCP) automated resting-state functional processing framework for newborn infants. *Neuroimage*. **223**, 117303 (2020).
78. Kuklisova-Murgasova, M., Quaghebeur, G., Rutherford, M. A., Hajnal, J. V. & Schnabel, J. A. Reconstruction of fetal brain MRI with intensity matching and complete outlier removal. *Med. Image Anal.* **16**, 1550–1564 (2012).
79. Makropoulos, A. et al. Automatic whole-brain MRI segmentation of the developing neonatal brain. *IEEE Trans. Med. Imaging*. **33**, 1818–1831 (2014).
80. Schuh, A. et al. A deformable model for the reconstruction of the neonatal cortex. In *Proc. International Symposium on Biomedical Imaging* 800–803 https://doi.org/10.1109/ISBI.2017.7950639 (2017).
81. Robinson, E. C. et al. Multimodal surface matching with higher-order smoothness constraints. *Neuroimage* **167**, 453–465 (2018).
82. Robinson, E. C. et al. MSM: a new flexible framework for Multimodal Surface Matching☆. *Neuroimage*. **100**, 414 (2014).
83. Bozek, J. et al. Construction of a neonatal cortical surface atlas using multimodal surface matching in the developing human connectome project. *Neuroimage*. **179**, 11–29 (2018).
84. Glasser, M. F. et al. The minimal preprocessing pipelines for the Human Connectome Project. *Neuroimage*. **80**, 105–124 (2013).
85. Ciric, R. et al. Benchmarking of participant-level confound regression strategies for the control of motion artifact in studies of functional connectivity. *Neuroimage*. **154**, 174–187 (2017).
86. Schaefer, A. et al. Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI. *Cereb. Cortex*. **28**, 3095–3114 (2018).
87. Zhi, D., King, M., Hernandez-Castillo, C. R. & Diedrichsen, J. Evaluating brain parcellations using the distance-controlled boundary coefficient. *Hum. Brain Mapp.* **43**, 3706–3720 (2022).
88. Shi, F. et al. Infant brain atlases from neonates to 1- and 2-year-olds. *PLoS ONE* **6**, e18746 (2011).
89. Seguin, C., Sporns, O. & Zalesky, A. Brain network communication: concepts, models and applications. *Nat. Rev. Neurosci.* **24**, 557–574 (2023).
90. Zamani Esfahlani, F., Faskowitz, J., Slack, J., Mišić, B. & Betzel, R. F. Local structure-function relationships in human brain networks across the lifespan. *Nat. Commun*. **13**, 2053 (2022).
91. Wood, S. Stable and efficient multiple smoothing parameter estimation for generalized additive models. *J. Am. Stat. Assoc.* **99**, 673–686 (2004).
92. Wood, S. N. Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *J. R. Stat. Soc. B Stat. Methodol.* **73**, 3–36 (2011).
93. Winkler, A. M., Ridgway, G. R., Webster, M. A., Smith, S. M. & Nichols, T. E. Permutation inference for the general linear model. *Neuroimage*. **92**, 381–397 (2014).
94. Smith, S. M. & Nichols, T. E. Threshold-free cluster enhancement: addressing problems of smoothing, threshold dependence and localisation in cluster inference. *Neuroimage*. **44**, 83–98 (2009).
95. Zöller, D. et al. Large-scale brain network dynamics provide a measure of psychosis and anxiety in 22q11.2 deletion syndrome. *Biol. Psychiatry Cogn. Neurosci. Neuroimaging*. **4**, 881–892 (2019).
96. Krishnan, A., Williams, L. J., McIntosh, A. R. & Abdi, H. Partial least squares (PLS) methods for neuroimaging: a tutorial and review. *Neuroimage*. **56**, 455–475 (2011).
97. Margulies, D. S. et al. Situating the default-mode network along a principal gradient of macroscale cortical organization. *Proc. Natl. Acad. Sci. USA*. **113**, 12574–12579 (2016).

Acknowledgements

This work was supported by the National Key Research and Development Program of China (2023YFF1204303), the National Natural Science Foundation of China (82202249, 82071994), the Science and Technology Support Program of Tianjin, China (25JCZDJC00210), Autonomous Project of Haihe Laboratory of Brain-Computer Interaction and Human-Machine Integration (25HHNJSS00010) and Tianjin Municipal Science & Technology Program (25ZXWZSY00060).

Author contributions

W. L.: Methodology, Formal Analysis, Writing Original Draft, Review & Editing; Visualization; Y. Chen: Conceptualization, Methodology, Supervision, Project Administration, Funding Acquisition; X. W.: Formal Analysis, Visualization; T. F.: Formal Analysis, Visualization; R. W.: Methodology, Visualization; Y. Cheng: Review & Editing; X. Z.: Supervision, Project Administration; Q. F.: Project Administration, Funding Acquisition; W. G.: Review & Editing; D. M: Conceptualization, Supervision, Project Administration, Funding Acquisition.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-70660-4.

Correspondence and requests for materials should be addressed to Yuanyuan Chen or Dong Ming.

Peer review information *Nature Communications* thanks Jingxin Nie and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026